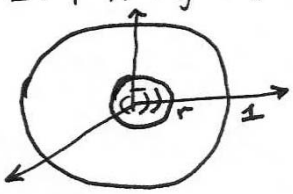
Chapter 6: Vector Random Variables

6.1 Vector Random Variables

6.1

a) $P[x^2+y^2+z^2 \le r \mid x^2+y^2+z^2 \le 1] = \frac{\frac{4\pi r^3}{3}}{\frac{4\pi 1^3}{3}} = r^3, \quad r \le 1$



b) $P[|X| \le \frac{1}{\sqrt{3}} \cap |Y| < \frac{1}{\sqrt{3}} \cap |Z| \le \frac{1}{\sqrt{3}} \mid x^2+y^2+z^2 \le 1]$

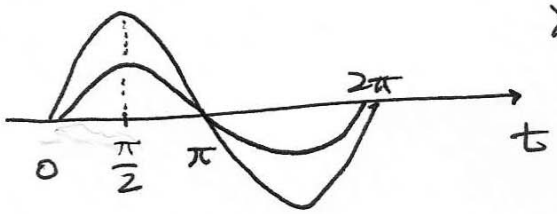
$$= \frac{\frac{2}{\sqrt{3}} \cdot \frac{2}{\sqrt{3}} \cdot \frac{2}{\sqrt{3}}}{\frac{4\pi 1^3}{3}} = \frac{8}{3\sqrt{3}4\pi} = \frac{2}{\pi\sqrt{3}} = 0.3675$$

c) $P[x>0 \cap y>0 \cap z>0] = \frac{1}{8}$

d) $P[z \le 0 \mid x^2+y^2+z^2 \le 1] = \frac{1}{2}$

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6.2



$X(t) = A\sin t$

A unif [0,1]

(a) $X_1 = X(0) = 0 \quad X_2 = X(\frac{\pi}{2}) = A \quad X_3 = X(\pi) = 0$

$$F_{\underline{X}}(x_1, x_2, x_3) = P[X_1 \le x_1, X_2 \le x_2, X_3 \le x_3]$$

$$= \begin{cases} 0 & \text{if } x_1 < 0 \text{ or } x_2 < 0 \text{ or } x_3 < 0 \\ P[A \le x_2] & x_1 > 0 \text{ and } x_3 > 0 \text{ and } x_2 > 0 \end{cases}$$

$X(t_1), X(t_2), X(t_3)$ are independent in this case

(b)

$X_1 = X(t_1)$	$X_2 = X(t_1 + \frac{\pi}{2})$	$X_3 = X(t_1 + \pi)$
$= A\sin t_1$	$= A\sin(t_1 + \frac{\pi}{2})$	$= A\sin(t_1 + \pi)$
$= \frac{1}{2}A$	$= A\cos t_1$	$= -A\sin t_1$
	$= \sqrt{3}/2$	$= -\frac{1}{2}A$

$$F_{\underline{X}}(x_1, x_2, x_3) = P[A\sin t_1 \le x_1, A\cos t_1 \le x_2, -A\sin t_1 \le x_3]$$

$$= P[A \le 2x_1, A \le \frac{2}{\sqrt{3}}x_2, A \ge 2x_3]$$

This is the prob. of 3 events involving A which is readily found once x_1, x_2 and x_3 are specified.

$X(t_1), X(t_2)$ and $X(t_3)$ are not independent.

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6.3

(a) $$P[|X|<5,\ Y<4,\ Z^3>8] = P[|X|<5]P[Y<4]P[Z^3>8]$$
$$= P[-5<X<5]P[Y<4]P[Z>2]$$
$$= [F_X(5^-) - F_X(-5)][F_Y(4^-)][1 - F_Z(2)]$$

(b) $$P[X=5,\ Y<0,\ Z>1] = P[X=5]P[Y<0]P[Z>1]$$
$$= [F_X(5) - F_X(5^-)]F_Y(0^-)[1 - F_Z(1)]$$

(c) $$P[\min(X,Y,Z)<2] = 1 - P[\min(X,Y,Z)\geq 2]$$
$$= 1 - P[X\geq 2]P[Y\geq 2]P[Z\geq 2]$$
$$= 1 - [1 - F_X(2^-)][1 - F_Y(2^-)][1 - F_Z(2^-)].$$

(d) $$P[\max(X,Y,Z)>6] = 1 - P[\max(X,Y,Z)\leq 6]$$
$$= 1 - P[X\leq 6]P[Y\leq 6]P[Z\leq 6]$$
$$= 1 - F_X(6)F_Y(6)F_Z(6).$$

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6.4

a) $$F_{\bar{X}}(x_1, x_2, x_3) = P[N_1 \le x_1 - s, N_2 \le x_2 - s, N_3 \le x_3 - s]$$
$$= F_{N_1}(x_1 - s) F_{N_2}(x_2 - s) F_{N_3}(x_3 - s)$$

$$f_{\bar{X}}(x_1, x_2, x_3) = f_{N_1}(x_1 - s) f_{N_2}(x_2 - s) f_{N_3}(x_3 - s)$$
$$= \frac{e^{-(x_1-s)^2/2}}{\sqrt{2\pi}} \frac{e^{-(x_2-s)^2/2}}{\sqrt{2\pi}} \frac{e^{-(x_3-s)^2/2}}{\sqrt{2\pi}}$$

b) $$P[\min(X_1, X_2, X_3) > 0] = P[X_1 > 0] P[X_2 > 0] P[X_3 > 0]$$
$$F_Y(y) = 1 - (1 - F_{N_1}(-s))(1 - F_{N_2}(-s))(1 - F_{N_3}(-s))$$
$$= 1 - (1 - F_N(-s))^3$$
$$= 1 - (1 - \Phi_N(-s))^3$$

c) $$P[X_1 > 0, X_2 > 0, X_3 > 0] + P[X_1 > 0, X_2 > 0, X_3 \le 0]$$
$$+ P[X_1 \le 0, X_2 > 0, X_3 > 0] + P[X_1 > 0, X_2 \le 0, X_3 > 0]$$
$$= (1 - F_N(-s))^3 + 3 F_N(-s)(1 - F_N(-s))^2$$

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6.5

$I_k = 1$ if kth drawn is black

1B, 2W

(a) $(I_1, I_2, I_3) \in \{000, 010, 001, 100, 011, 101, 110, 111\} = \mathcal{S}$

$p_{\underline{I}}(ijk) = \frac{1}{8}$ all $ijk \in \mathcal{S}$

$X = I_1 + I_2 + I_3$ $\quad X \in \{0, 1, 2, 3\}$

$Y = \min(I_1, I_2, I_3)$ $\quad Y \in \{0, 1\}$

$Z = \max(I_1, I_2, I_3)$ $\quad Z \in \{0, 1\}$

$\underline{I}$	X	Y	Z
000	0	0	0
001	1	0	1
010	1	0	1
100	1	0	1
011	2	0	1
101	2	0	1
110	2	0	1
111	3	1	1

$P_{\underline{X}}[000] = 1/8$

$P_{\underline{X}}[101] = 3/8$

$P_{\underline{X}}[201] = 3/8$

$P_{\underline{X}}[311] = 1/8$

$P[X=0] = 1/8$

$P[X=1] = 3/8$

$P[X=2] = 3/8$

$P[X=3] = 1/8$

$P[Y=0] = 7/8$

$P[Y=1] = 1/8$

$P[Z=0] = 1/8$

$P[Z=1] = 7/8$

(b) X and Z not indep since $P[Y=1, Z=1] = \frac{1}{8} \neq P[Y=1]P[Z=1]$ ($\frac{1}{8}$, $\frac{7}{8}$)

X and Y not indep since $P[X=0, Y=0] = \frac{1}{8} \neq P[X=0]P[Y=0]$

X and Z not indep since $P[X=0, Z=0] \neq P[X=0]P[Z=0]$.

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(6.5) — continued —

(c) $I \in \{001, 010, 100\}$

I	X	Y	Z
001	1	0	1
010	1	0	1
100	1	0	1

$P[101] = 1$

$P[X=1] = 1$
$P[Y=0] = 1$
$P[Z=1] = 1$

$P[X=1, Y=0] = P[X=1]P[Y=0] = 1$
$P[X=0, Y=0] = P[X=0]P[Y=0] = 0$
etc
} X and Y are indep.

Similarly X and Z are independent
Y and Z are independent

$P[X=1, Y=0, Z=1] = P[X=1]P[Y=0]P[Z=1] = 1$
all other triplets have prob zero

∴ X, Y, Z are independent

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6.6

Every time instant there are 3 arrivals; of these

(a) $X_1 = i$ go to port 1, $X_2 = j$ to port 2, $X_3 = k$ to port 3

and $3-i-j-k$ to the fictional port #4

$$P[X_1 = i, X_2 = j, X_3 = k] = \frac{3!}{i!\,j!\,k!\,(3-i-j-k)!}\left(\frac{p}{3}\right)^i\left(\frac{p}{3}\right)^j\left(\frac{p}{3}\right)^k(1-p)^{3-i-j}$$

(b) $$P[X_1 = i, X_2 = j] = \sum_{k=0}^{3-i-j} P[X_1 = i, X_2 = j, X_3 = k] = \ldots$$

Alternatively, $N_1 = i$ go to port 1, $N_2 = j$ to port 2

and $3-i-j$ to ports 3 or 4 (prob $1-2\frac{p}{3}$)

$$P[X_1 = i, X_2 = j] = \frac{3!}{i!\,j!\,(3-i-j)!}\left(\frac{p}{3}\right)^i\left(\frac{p}{3}\right)^j\left(1-2\frac{p}{3}\right)^{3-i-j}$$

$i \geq 0 \quad j \geq 0 \quad i+j \leq 3$

(c) $$P[X_2 = j] = \sum_{i=0}^{3-j} P[X_1 = i, X_2 = j] = \ldots$$

or reasoning as above

$$P[X_2 = j] = \frac{3!}{j!\,(3-j)!}\left(\frac{p}{3}\right)^j\left(1-\frac{p}{3}\right)^{3-j}$$

(d) Consider $P[X_1 = 0, X_2 = 0] = \left(1-\frac{p}{3}\right)^3$ $\neq$ Not Independent

$$P[X_1 = 0]P[X_2 = 0] = \left(1-\frac{p}{3}\right)^3\left(1-\frac{p}{3}\right)^3$$

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6.6

(a) $(X_i - 1)^+ =$ # pkts discarded by port i.

$$E[(X_i-1)^+] = 1 \cdot P[X_i = 2] + 2P[X_i = 3]$$

$$= 1 \cdot \frac{3!}{2!1!}\left(\frac{p}{3}\right)^2\left(1-\frac{p}{3}\right)^1 + 2\left(\frac{p}{3}\right)^3$$

$$= 3\left(\frac{p}{3}\right)^2\left(1-\frac{p}{3}\right) + 2\left(\frac{p}{3}\right)^3$$

Average # discarded by all ports is

$$3 \cdot E[(X_i-1)^+] = 9\left(\frac{p}{3}\right)^2\left(1-\frac{p}{3}\right) + 2\left(\frac{p}{3}\right)^3$$

6.7 a)

$$\begin{aligned} 1 &= \int_0^1\int_0^1\int_0^1 k(x+y+z)dxdydz \\ &= k\int_0^1\int_0^1\left(\frac{1}{2}+y+z\right)dydz \\ &= k\int_0^1\left(\left(\frac{1}{2}+z\right)+\frac{1}{2}\right)dz \\ &= k\left(1+\frac{1}{2}\right) \Rightarrow k = \frac{2}{3} \end{aligned}$$

b) $f_{XY}(x,y) = \frac{2}{3}\int_0^1 (x+y+z)dz = \frac{2}{3}\left[x+y+\frac{1}{2}\right]$

$$f_Z(z|x,y) = \frac{f_{XYZ}(x,y,z)}{f_{XY}(x,y)} = \frac{x+y+z}{x+y+\frac{1}{2}}$$

c) $f_X(x) = \frac{2}{3}\int_0^1 (x+y+\frac{1}{2})\,dy = \frac{2}{3}\left[xy\Big|_0^1 + \frac{y^2}{2}\Big|_0^1 + \frac{1}{2}y\Big|_0^1\right] = \frac{2}{3}\left[x+1\right]$.

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6.8

~~40~~ a) $$f_{X,Y}(x,y) = \int_{-\infty}^{\infty} f_{X,Y,Z}(x,y,z')dz'$$
$$= \int_{-(1-x^2-y^2)^{1/2}}^{(1-x^2-y^2)^{1/2}} \frac{3}{4\pi} dz'$$
$$= \frac{3}{2\pi}(1-x^2-y^2)^{1/2}$$

b) $$f_X(x) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y,Z}(x,y',z')dy'dz'$$
$$= \int_{-(1-x^2)^{1/2}}^{(1-x^2)^{1/2}} \int_{(1-x^2-y'^2)^{1/2}}^{(1-x^2-y'^2)^{1/2}} \frac{3}{4\pi} dz'dy'$$
$$= \int_{-(1-x^2)^{1/2}}^{(1-x^2)^{1/2}} \frac{3}{2\pi}(1-x^2-y'^2)^{1/2}dy'$$

Let $a^2 = 1-x^2$:

$$\int_0^a \sqrt{a^2-t^2}\, dt = \int_0^{\pi/2} a\cos u\, a\cos u du \qquad (= a\sin u)$$
$$= \frac{1}{2}a^2\int_0^{\pi/2}(1+\cos 2u)du$$
$$= \frac{1}{2}a^2\left(\frac{\pi}{2} + \frac{1}{2}\sin 2u|_0^{\pi/2}\right)$$
$$= \frac{1}{4}\pi a^2$$

$$\therefore \quad f_X(x) = \frac{3}{2\pi}\cdot 2\cdot\frac{1}{4}\pi a^2 = \frac{3}{4}(1-x^2)$$

c) $$f(x,y|z) = f(x,y,z)/f(z)$$
$$= \frac{\frac{3}{4\pi}}{\frac{3}{4}(1-z^2)}$$
$$= \frac{1}{\pi(1-z^2)}$$

d) X, Y, Z are not independent RVs.

e) $P[A] = P[R > \frac{1}{2}, \text{ in positive orthant}] = \frac{1}{8}\left(1 - \frac{4\pi r^3}{4\pi}\right) = \frac{1}{8}(1-r^3)$

$$f(x,y,z|A) = \frac{3}{32\pi(1-r^3)}$$

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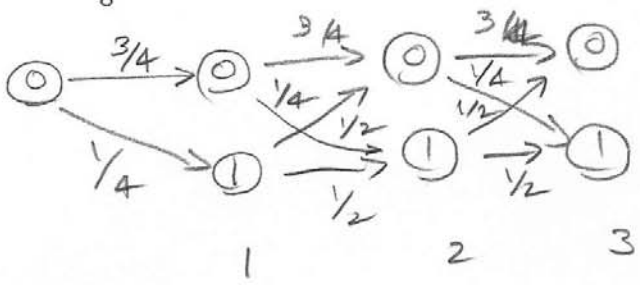
6.9

$$p_{X_1X_2X_3}(x_1,x_2,x_3) = p_{X_3}(x_3|x_1,x_2)\,p_{X_1X_2}(x_1,x_2)$$

$$= p_{X_3}(x_3|x_1,x_2)\,p_{X_2}(x_2|x_1)\,p_X(x_1).$$

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6.10 $X_0 = 0$



$$P[X_3X_2X_1 = 000] = P[X_3=0|00]P[X_2=0|X_1=0]P[X_1=0]$$
$$= \frac{3}{4}\frac{3}{4}\frac{3}{4}$$

$$P[100] = P[X_3=1|00]P[0|0]P[0]$$
$$\frac{1}{4} \quad \frac{3}{4} \quad \frac{3}{4}$$

$$P[010] = P[X_3=0|10]P[1|0]P[0]$$
$$\frac{1}{2} \quad \frac{1}{4} \quad \frac{3}{4}$$

$$P[110] = P[X_3=1|10]P[1|0]P[0]$$
$$\frac{1}{2} \quad \frac{1}{4} \quad \frac{3}{4}$$

$$P[111] = P[X_3=1|11]P[1|1]P[1]$$
$$\frac{1}{2} \quad \frac{1}{2} \quad \frac{1}{4}$$

$$P[011] = P[X_3=0|11]P[1|1]P[1]$$
$$\frac{1}{2} \quad \frac{1}{2} \quad \frac{1}{4}$$

$$P[101] = P[X_3=1|01]P[0|1]P[1]$$
$$\frac{1}{4} \quad \frac{1}{2} \quad \frac{1}{4}$$

$$P[001] = P[X_3=0|01]P[0|1]P[1]$$
$$\frac{3}{4} \quad \frac{1}{2} \quad \frac{1}{4}$$

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6.10 continued

X_3 X_2 X_1	Prob
0 0 0	27/64
1 0 0	9/64
0 1 0	6/64
1 1 0	6/64
0 1 1	4/64
1 0 1	4/64
0 0 1	2/64
1 1 1	6/64

$$P[X_3=0] = \frac{27+6+4+6}{64} = \frac{43}{64}$$

$$P[X_3=1] = \frac{21}{64}$$

$$P[X_2=0] = \frac{27+9+2+6}{64} = \frac{44}{64}$$

$$P[X_2=1] = \frac{20}{64}$$

$$P[X_1=0] = \frac{27+9+6+6}{64} = \frac{48}{64}$$

$$P[X_1=1] = \frac{16}{64}$$

6.11

$$\begin{aligned} f_{XYZ}(x,y,z) &= f_Z(z|x,y)f_{XY}(x,y) \\ &= f_Z(z|x,y)f_Y(y|x)f_X(x) \end{aligned}$$

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6.12

a)
$$\begin{aligned} f_Y(y|x) &= f_{U_2}(y-x) \\ f_Z(z|x,y) &= f_{U_3}(z-y) \\ f_{X,Y,Z}(x,y,z) &= f_Z(z|x,y)f_Y(y|x)f_X(x) \\ &= f_{U_3}(z-y)f_{U_2}(y-x)f_{U_1}(x) \end{aligned}$$

b)
$$\begin{aligned} f_{YZ}(y,z) &= \int_{-\infty}^{\infty} f_{U_3}(z-y)f_{U_2}(y-x)f_{U_1}(x)dx \\ &= f_{U_3}(z-y)\underbrace{\int_{-\infty}^{\infty} f_{U_2}(y-x)f_{U_1}(x)dx}_{f_Y(y)} \end{aligned}$$

We next find $f_Y(y)$:
For $0 \le y \le 1$

$$\begin{aligned} f_Y(y) &= \int_0^y f_{U_1}(u_1)\cdot f_{U_2}(y-u_1)du_1 \\ &= \int_0^y 1\cdot 1 du_1 \\ &= y \end{aligned}$$

For $1 \le y \le 2$

$$\begin{aligned} f_Y(y) &= \int_{y-1}^1 f_{U_1}(u_1)f_{U_2}(y-u_1)du_1 \\ &= 2-y \end{aligned}$$

$$\therefore \quad f_{Y,Z}(y,z) = \begin{cases} y & 0 \le y \le 1, \quad y \le z \le y+1 \\ 2-y & 1 \le y \le 2, \quad y \le z \le y+1 \\ 0 & \text{elsewhere} \end{cases}$$

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The pdf of Z is:

$$f_Z(z) = \int f_{Y,Z}(y', z)dy'$$

For $0 \leq z \leq 1$

$$f_Z(z) = \int_0^z y'dy' = \frac{1}{2}z^2$$

For $1 \leq z \leq 2$

$$\begin{aligned} f_Z(z) &= \int_{z-1}^1 y'dy' + \int_1^z (2-y')dy' \\ &= \frac{1}{2}[1-(z-1)^2] + 2(z-1) - \frac{1}{2}[z^2-1] \\ &= \frac{1}{2}[1-z^2+2z-1] + 2z - 2 - \frac{1}{2}[z^2-1] \\ &= -z^2 + 3z - \frac{3}{2} \end{aligned}$$

For $2 \leq z \leq 3$

$$\begin{aligned} f_Z(z) &= \int_{z-1}^2 (2-y')dy' \\ &= 2[2-(z-1)] - \frac{1}{2}[4-(z-1)^2] \\ &= 6 - 2z - \frac{1}{2}[-z^2+2z+3] \\ &= \frac{1}{2}z^2 - 3z + \frac{9}{2} \\ f_Z(z) &= \begin{cases} \frac{1}{2}z^2 & 0 \leq z \leq 1 \\ -z^2+3z-\frac{3}{2} & 1 \leq z \leq 2 \\ \frac{1}{2}z^2-3z+\frac{9}{2} & 2 \leq z \leq 3 \end{cases} \end{aligned}$$

c) From part b) we have

$$\begin{aligned} f_{Y,Z}(y,z) &= f_{U_3}(z-y)f_Y(y) \\ f_{U_3}(z-y) &= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{(z-y)^2}{2}\right) \\ f_Y(y) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\{(y-x)^2+x^2\}}dx = \frac{1}{\sqrt{2\pi\cdot 2}}e^{-\frac{y^2}{2\cdot 2}} \end{aligned}$$

$$\begin{aligned} f_{Y,Z}(y,z) &= \frac{1}{2\pi\sqrt{2}}\exp\left[-\frac{y^2}{4} - \frac{(z-y)^2}{2}\right] \\ f_Z(z) &= \frac{1}{2\pi\sqrt{2}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\{(z-y)^2+\frac{y^2}{2}\}}dy \\ &= \frac{1}{\sqrt{2\pi\cdot 3}}\exp\left(-\frac{y^2}{2\cdot 3}\right) \end{aligned}$$

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6.13

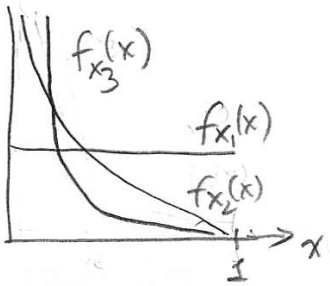
Example 6.7

a) Using the result from ~~problem 41~~:

$$f_{X_1X_2X_3} = f_{X_1}(x_1)f_{X_2}(x_2|x_1)f_{X_3}(x_3|x_1,x_2)$$
$$= 1 \cdot \frac{1}{x_1} \cdot \frac{1}{x_2} \quad \text{for } 0 < x_3 < x_2 < x_1 < 1.$$

b) Here we need to carefully determine the limits of the integrals: For a given x_3, x_2 varies from x_3 to 1; for a given x_2, x_1 varies from x_2 to 1. Thus

$$f_{X_3}(x_3) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X_1X_2X_3}(x_1,x_2,x_3)dx_1dx_2$$
$$= \int_{x_3}^{1} dx_2 \int_{x_2}^{1} \frac{dx_1}{x_1x_2} = \int_{x_3}^{1} \frac{dx_2}{x_2} \ln x_1|_{x_2}^{1}$$
$$= \int_{x_3}^{1} (-1)\frac{\ln x_2}{x_2}dx_2 = (-1)\frac{(\ln x_2)^2}{2}\Big|_{x_3}^{1}$$
$$= \frac{(-1)^2}{2}(\ln x_3)^2$$



$$f_{X_2}(x_2) = \int_0^{x_2} dx_3 \int_{x_2}^{1} \frac{dx_1}{x_1x_2} = -\int_0^{x_2} dx_3 \frac{\ln x_2}{x_2} = -\ln x_2$$

We could also find the marginal pdf of X_2 by noting from the way the experiment is defined that:

$$f_{X_1X_2}(x_1,x_2) = 1 \cdot \frac{1}{x_1} \qquad 0 < x_2 < x_1 < 1$$

Thus

$$f_{X_2}(x_2) = \int_{x_2}^{1} f_{X_1X_2}(x_1,x_2)dx_1 = \int_{x_2}^{1} \frac{dx_1}{x_1} = -\ln x_2 \ .$$

Clearly X_1 is uniform in [0,1]. Nevertheless we carry out the integral to find $f_{X_1}(x_1)$:

$$f_{X_1}(x_1) = \int_0^{x_1} dx_2 \int_0^{x_2} \frac{dx_3}{x_1x_2}$$
$$= \int_0^{x_1} dx_2 \frac{1}{x_1} = 1 \qquad 0 < x_1 < 1 \ .$$

b)
$$f_{X_3}(x_3|x_1) = \int_{x_3}^{x_1} f_{X_2X_3}(x_2,x_3|x_1)dx_2$$
$$= \int_{x_3}^{x_1} \frac{f(x_1,x_2,x_3)}{f(x_1)}dx_2$$
$$= \int_{x_3}^{x_1} \frac{dx_2}{x_1x_2}$$
$$= \frac{1}{x_1}[\ln x_1 - \ln x_3] = \frac{1}{x_1}\ln\frac{x_1}{x_3} \qquad 0 < x_3 < x_1$$

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3) (c)

$$f(x_1, x_2 | x_3) = \frac{1}{x_1 x_2} \frac{2}{(\ln x_3)^3} \qquad 0 < x_3 < x_2 < x_1 < 1$$

$$f(x_1 | x_3) = \int_{x_3}^{x_1} \frac{2}{x_1 (\ln x_3)^2} \frac{1}{x_2} dx_2 = \frac{2}{x_1 (\ln x_3)^2} \left[\ln x_1 - \ln x_3 \right]$$

$$= \frac{2 \ln \frac{x_1}{x_3}}{x_1 (\ln x_3)^2} \qquad x_3 < x_1 < 1$$

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6.14

a) $$p(X_1=x_1, X_2=x_2, X_3=x_3, X_4=x_4) = \frac{n!}{x_1!\,x_2!\,x_3!\,x_4!}\left(\frac{1}{2}\right)^{x_1}\left(\frac{1}{4}\right)^{x_2}\left(\frac{1}{8}\right)^{x_3}\left(\frac{1}{8}\right)^{x_4}$$

$$p(X_1=x_1, X_2=x_2, X_3=x_3) = \frac{n!}{x_1!\,x_2!\,x_3!\,(n-x_1-x_2-x_3)!}\left(\frac{1}{2}\right)^{x_1}\left(\frac{1}{4}\right)^{x_2}\left(\frac{1}{8}\right)^{x_3}\left(\frac{1}{8}\right)^{n-x_1-x_2-x_3}$$

b) $$p(x_1, x_2) = \sum_{x_3=0}^{n} \frac{n!}{x_1!\,x_2!\,x_3!\,(n-x_1-x_2-x_3)!}\left(\frac{1}{2}\right)^{x_1}\left(\frac{1}{4}\right)^{x_2}\left(\frac{1}{8}\right)^{x_3}\left(\frac{1}{8}\right)^{n-x_1-x_2-x_3}$$

$$= \frac{\left(\frac{1}{2}\right)^{x_1}\left(\frac{1}{4}\right)^{x_2}}{x_1!\,x_2!}\sum_{x_3=0}^{n}\frac{(n-x_1-x_2)!}{x_3!\,(n-x_1-x_2-x_3)!}\frac{n!}{(n-x_1-x_2)!}\left(\frac{1}{8}\right)^{x_3}\left(\frac{1}{8}\right)^{n-x_1-x_2-x_3}$$

$$= \frac{\left(\frac{1}{2}\right)^{x_1}\left(\frac{1}{4}\right)^{x_2}}{x_1!\,x_2!}\frac{n!}{(n-x_1-x_2)!}\left(\frac{1}{8}+\frac{1}{8}\right)^{n-x_1-x_2}$$

c) $$p(x_1) = \sum_{x_2=0}^{n}\left(\frac{1}{2}\right)^{x_1}\left(\frac{1}{4}\right)^{x_2}\left(\frac{1}{4}\right)^{n-x_1-x_2}\frac{n!}{x_1!\,x_2!\,(n-x_1-x_2)!}$$

$$= \frac{\left(\frac{1}{2}\right)^{x_1}}{x_1!}\sum_{x_2=0}^{n}\frac{n!}{(n-x_1)!}\cdot\frac{(n-x_1)!}{(n-x_1-x_2)!\,x_2!}\left(\frac{1}{4}\right)^{x_2}\left(\frac{1}{4}\right)^{n-x_1-x_2}$$

$$= \frac{\left(\frac{1}{2}\right)^{x_1}}{x_1!}\frac{n!}{(n-x_1)!}\left(\frac{1}{2}\right)^{n-x_1}$$

$$= \binom{n}{x_1}\left(\frac{1}{2}\right)^{n}$$

d) $$p(x_2, x_3 \mid x_1=m) = \frac{\frac{n!}{m!\,x_2!\,x_3!\,(n-x_2-x_3-m)!}\left(\frac{1}{2}\right)^{m}\left(\frac{1}{4}\right)^{x_2}\left(\frac{1}{8}\right)^{x_3}\left(\frac{1}{8}\right)^{n-m-x_2-x_3}}{\frac{n!}{m!\,(n-m)!}\left(\frac{1}{2}\right)^{n}}$$

$$= \frac{(n-m)!}{x_2!\,x_3!\,(n-m-x_2-x_3)!}\underbrace{\left(\frac{1}{2}\right)^{m}\left(\frac{1}{2}\right)^{-n}}_{\left(\frac{1}{4}\right)^{x_2}\left(\frac{2}{8}\right)^{x_3}\left(\frac{2}{8}\right)^{n-m-x_2-x_3}}\left(\frac{1}{4}\right)^{x_2}\left(\frac{1}{8}\right)^{x_3}\left(\frac{1}{8}\right)^{n-m-x_2-x_3}$$

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6.15

$$p_N(n) = \frac{\alpha^n}{n!} e^{-\alpha} \qquad n = 0, 1, \ldots$$

$$P_{\vec{X}}(x_1, x_2, x_3, x_4) = P(x_1, x_2, x_3, x_4 \mid n)\, p_N(n)$$

$$= \frac{n!}{x_1! x_2! x_3! x_4!} \left(\frac{1}{2}\right)^{x_1} \left(\frac{1}{4}\right)^{x_2} \left(\frac{1}{8}\right)^{x_3} \left(\frac{1}{8}\right)^{x_4} \frac{\alpha^n e^{-\alpha}}{n!}$$

$$n = x_1 + x_2 + x_3 + x_4$$

$$= \frac{1}{x_1! x_2! x_3! x_4!} \left(\frac{\alpha}{2}\right)^{x_1} \left(\frac{\alpha}{4}\right)^{x_2} \left(\frac{\alpha}{8}\right)^{x_3} \left(\frac{\alpha}{8}\right)^{x_4} e^{-\alpha}$$

$$= \left(\frac{\left(\frac{\alpha_1}{2}\right)^{x_1}}{x_1!} e^{-\frac{\alpha_1}{2}}\right) \left(\frac{\left(\frac{\alpha_1}{4}\right)^{x_2}}{x_2!} e^{-\frac{\alpha}{4}}\right) \left(\frac{\left(\frac{\alpha}{8}\right)^{x_3}}{x_3!} e^{-\frac{\alpha}{8}}\right) \left(\frac{\left(\frac{\alpha}{8}\right)^{x_4}}{x_4!} e^{-\frac{\alpha}{8}}\right)$$

independent Poisson RV's !

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6.16

~~4.46~~ For $k_j \geq 0$ such that $k_1 + k_2 + k_3 \leq$n

$$p(k_1, k_2, k_3) = \frac{1}{\binom{n+3}{3}}$$

Note: $\binom{n+3}{3}$ is the number of ways of distributing n identical balls in 4 cells: See Sampling with Replacement and Without Ordering in Section 2.3.

a) $$p(k_1, k_2) = \sum_{k_3=0}^{n-k_1-k_2} p(k_1, k_2, k_3) = \frac{n - k_1 - k_2 + 1}{\binom{n+3}{3}}$$

b) $$p(k_1) = \sum_{k_2=0}^{n-k_1} \frac{n - k_1 - k_2 + 1}{\binom{n+3}{3}} \qquad j = n - k_1 - k_2 + 1$$

$$= \sum_{j=1}^{n-k_1+1} \frac{j}{\binom{n+3}{3}} = \frac{(n - k_1 + 2)(n - k_1 + 1)}{2\binom{n+3}{3}}$$

Check

$$\sum_{k_1=0}^{n} p(k_1) = \frac{1}{2\binom{n+3}{3}} \sum_{k_1=0}^{n} (n - k_1 + 2)(n - k_1 + 1) \qquad j = n - k_1 + 1$$

$$= \frac{1}{2\binom{n+3}{3}} \sum_{j=1}^{n+1} j(j+1) \qquad \sum_{j=1}^{n+1} j = \frac{(n+2)(n+1)}{2}, \quad \sum_{j=1}^{n+1} j^2 = \frac{(n+1)(n+2)(2n+3)}{6}$$

$$= \frac{1}{2\binom{n+3}{3}} \underbrace{\left[\frac{(n+2)(n+1)}{2} + \frac{(n+1)(n+2)(2n+3)}{6}\right]}$$

$$= \frac{1}{2\binom{n+3}{3}} \frac{(n+1)(n+2)(n+3)}{3} = 1 \quad \surd$$

c) $$p(k_2, k_3 | k_1) = \frac{p(k_1, k_2, k_3)}{p(k_1)} = \frac{\dfrac{(n - k_1 + 2)(n - k_1 + 1)}{2\binom{n+3}{3}}}{\binom{n+3}{3}}$$

$$= \frac{(n - K - 1 + 2)(n - k_1 + 1)}{2}$$

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6.17

$j \geq 0, k \geq 0, l \geq 0$

$$P[N_1=j, N_2=k, N_3=l \mid T=t] = \frac{(\lambda_1 t)^j}{j!}\frac{(\lambda_2 t)^k}{k!}\frac{(\lambda_3 t)^l}{l!} e^{-(\lambda_1+\lambda_2+\lambda_3)t}$$

$$P[N_1=j, N_2=k, N_3=l] = \int_0^\infty \frac{\lambda_1^j \lambda_2^k \lambda_3^l}{j!k!l!} t^{j+k+l} e^{-(\lambda_1+\lambda_2+\lambda_3)t} \alpha e^{-\alpha t} dt$$

$$= \frac{\alpha \lambda_1^j \lambda_2^k \lambda_3^l}{j!k!l!} \int_0^\infty t^{j+k+l} e^{-(\alpha+\lambda_1+\lambda_2+\lambda_3)t} dt$$

$$\frac{\Gamma(j+k+l+1)}{(\alpha+\lambda_1+\lambda_2+\lambda_3)^{j+k+l+1}}$$

$$= \frac{(j+k+l)!}{j!k!l!}\left(\frac{\alpha}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)\left(\frac{\lambda_1}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)^j \left(\frac{\lambda_2}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)^k \left(\frac{\lambda_3}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)^l$$

Neat Answer!

Imagine a sequence of 4 way races among:

- customer w service : rate α wins w. prob $\frac{\alpha}{\alpha+\lambda_1+\lambda_2+\lambda_3}$
- arrival type 1 : rate λ_1 " $\frac{\lambda_1}{\alpha+\lambda_1+\lambda_2+\lambda_3}$
- " 2 " λ_2 " $\frac{\lambda_2}{\alpha+\lambda_1+\lambda_2+\lambda_3}$
- " 3 " λ_3 " $\frac{\lambda_3}{\alpha+\lambda_1+\lambda_2+\lambda_3}$

Game ends when customer w service wins.

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(6.17)

(b) $$P[N_1 = j] = \int_0^\infty \frac{(\lambda_1 t)^j}{j!} e^{-\lambda t} \alpha e^{-\alpha t} dt$$

$$= \frac{\alpha \lambda_1^j}{j!} \int_0^\infty t^j e^{-(\lambda_1+\alpha)t} dt$$

$$\int_0^\infty t^j e^{-(\lambda_1+\alpha)t} dt = \frac{\Gamma(j+1)}{(\alpha+\lambda_1)^{j+1}}$$

$$= \left(\frac{\alpha}{\alpha+\lambda_1}\right)\left(\frac{\lambda_1}{\alpha+\lambda_1}\right)^j \qquad j = 0, 1, \ldots$$

(c)

$$P[N_1 = j, N_2 = k \mid N_3 = l] =$$

$$= \frac{(j+k+l)!}{j!k!l!} \frac{\left(\frac{\alpha}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)\left(\frac{\lambda_1}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)^j \left(\frac{\lambda_2}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)^k \left(\frac{\lambda_3}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)^l}{\left(\frac{\alpha}{\alpha+\lambda_3}\right)\left(\frac{\lambda_3}{\alpha+\lambda_3}\right)^l}$$

$$= \frac{(j+k+l)!}{j!k!l!} \left(\frac{\alpha+\lambda_3}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)\left(\frac{\alpha+\lambda_3}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)^l \left(\frac{\lambda_1}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)^j \left(\frac{\lambda_2}{\alpha+\lambda_1+\lambda_2+\lambda_3}\right)^k$$

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6.2 Functions of Several Random Variables

6.18

a) $Y = \min(L_1, L_2, \ldots, L_N)$

$1 - F_Y(y) = P[\min(L_1, L_2, \ldots, L_N) > y]$

$F_Y(y) = 1 - (1 - F_L(y))^N$

$= 1 - \left(1 - \left(1 - \left(\frac{y_{min}}{y}\right)^{\alpha}\right)\right)^N$

$= 1 - \left(\frac{y_{min}}{y}\right)^{\alpha N}$

b) $F_Y(y) = 1 - \left(1 - \left(1 - e^{-(y/\lambda)^{\beta}}\right)\right)^N$

$= 1 - \left(e^{-(y/\lambda)^{\beta}}\right)^N$

6.19

$I_k(t) = 1$ iff item k still working.

$P[I_k(t) = 1] = P[X_k > t] = R(t)$ This is a Bernoulli R Trial success prob.

$N(t) = I_1(t) + \cdots + I_N(t)$ # successes in N Bernoulli trials

$N(t)$ is a Binomial RV

$P[N(t) = n] = \binom{N}{n} R(t)^n (1 - R(t))^{N-n}$

$E[N(t)] = N \cdot R(t)$

$VAR[N(t)] = N R(t)(1 - R(t))$

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6.20

$$P[\max(X_1^2, X_2^2, \ldots X_n^2) \le \gamma]$$

$$= P[X_1^2 \le \gamma] \ldots P[X_n^2 \le \gamma]$$

$$P[X^2 \le \gamma] = P[X \le \sqrt{\gamma}] = \int_0^{\sqrt{\gamma}} \frac{r}{\alpha^2} e^{-r^2/2\alpha^2} dr$$

$$= 1 - e^{-\gamma/2\alpha^2}$$

$$\therefore P[\max(\ ) \le \gamma] = \left(1 - e^{-\gamma/2\alpha^2}\right)^n$$

6.21

a) $$P_{Y_1 Y_2 \ldots Y_K}(y_1, y_2, \ldots y_K) = P[N_1 \le y_1 - \alpha_1 b_1, N_2 \le y_2 - \alpha_2 b_2, \ldots N_K \le y_K - \alpha_K b_K]$$

$$= F_{N_1 N_2 \ldots N_K}(y_1 - \alpha_1 b_1, y_2 - \alpha_2 b_2 \ldots y_K - \alpha_K b_K)$$

$$f_{\bar{Y}}(y_1, y_2, \ldots, y_K) = f_{N_1}(y_1 - \alpha_1 b_1) f_{N_2}(y_2 - \alpha_2 b_2) \ldots f_{N_K}(y_K - \alpha_K b_K)$$

$$= \left(\frac{1}{\sqrt{2\pi}}\right)^K e^{-(y_1 - \alpha_1 b_1)^2/2} e^{-(y_2 - \alpha_2 b_2)^2/2} \ldots e^{-(y_K - \alpha_K b_K)^2/2}$$

b) $$P[Y_1 > 0, Y_2 > 0, Y_3 > 0 \ldots Y_K > 0 \mid b_1 = 1, b_2 = 1, \ldots b_K = 1]$$

$$= P[N_1 > -\alpha_1 b_1, N_2 > -\alpha_2 b_2, \ldots N_K > -\alpha_K b_K \mid b_1 = 1, b = 2 \ldots b_K = 1]$$

$$= P[N_1 > -\alpha_1, N_2 > -\alpha_2 \ldots N_K > -\alpha_K]$$

$$= (1 - \Phi_{N_1}(-\alpha_1))(1 - \Phi_{N_2}(-\alpha_2)) \ldots (1 - \Phi_{N_K}(-\alpha_K))$$

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6.22
~~4.53~~ a) $\underline{Z} = \begin{bmatrix} U \\ V \\ W \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}_{A} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \qquad |A| = 1$

$$\begin{aligned} X_1 &= = U \\ X_2 &= V - X_1 = V - U \\ X_3 &= W - X_1 - X_2 = W - V \end{aligned}$$

$$f_{\underline{Z}}(u,v,w) = \left.\frac{f_{\underline{X}}(\underline{x})}{|A|}\right|_{\underline{x}=A^{-1}\underline{u}} = f_{\underline{X}}(u, v-u, w-v)$$

b) $$\begin{aligned} f_{\underline{Z}}(u,v,w) &= \frac{1}{(\sqrt{2\pi})^3} e^{-\frac{u^2}{2}} e^{-(v-u)^2/2} e^{-(w-v)^2/2} \\ &= \frac{1}{(\sqrt{2\pi})^3} e^{-\frac{1}{2}[2u^2+2v^2+w^2-2uv-2vw]} \\ &= \frac{1}{(\sqrt{2\pi})^3} e^{-[u^2+v^2+\frac{1}{2}w^2-uv-vw]} \end{aligned}$$

6.22 c)

$$f(v,w) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-[u^2-uv]} du \; e^{-[v^2+\frac{1}{2}w^2-vw]}$$

$$= e^{\frac{1}{4}v^2} \int_{-\infty}^{\infty} \frac{1/2}{\sqrt{2\pi}\,\frac{1}{2}} e^{-[u-\frac{1}{2}v]^2}$$

$$f(v,w) = \frac{1}{2\pi} e^{-[\frac{3}{4}v^2+\frac{1}{2}w^2-vw]}$$

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6.23 ~~4.34~~

$$M = \frac{1}{2}X_1 + \frac{1}{2}X_2 = \frac{1}{2}(X_1 + X_2)$$

$$V = \frac{1}{2}(X_1 - M)^2 + \frac{1}{2}(X_2^2 - M)^2 = \frac{1}{2}\left(\frac{X_1}{2} - \frac{X_2}{2}\right)^2 + \frac{1}{2}\left(\frac{X_2}{2} - \frac{X_1}{2}\right)^2$$

$$= \frac{1}{8}(X_1 - X_2)^2$$

(a)

$$\left.\begin{array}{l} \sqrt{8V} = X_1 - X_2 \\ 2M = X_1 + X_2 \end{array}\right\} \quad \begin{array}{l} X_1 = M + \sqrt{2V} \\ X_2 = M - \sqrt{2V} \end{array}$$

$$\left|\begin{array}{cc} \frac{\partial M}{\partial x_1} & \frac{\partial M}{\partial x_2} \\ \frac{\partial V}{\partial x_1} & \frac{\partial V}{\partial x_2} \end{array}\right| = \left|\begin{array}{cc} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4}(x_1 - x_2) & -\frac{1}{4}(x_1 - x_2) \end{array}\right| = \left|-\frac{1}{8}(x_1 - x_2) - \frac{1}{8}(x_1 - x_2)\right|$$

$$= \frac{1}{4}|x_1 - x_2| = \frac{1}{4}|2\sqrt{2V}| = \sqrt{v/2}$$

$$f_{M,V}(m,v) = \frac{f_{X_1,X_2}(m + \sqrt{2v}, m - \sqrt{2v})}{\sqrt{v/2}}$$

(c)

If $f_{X_1X_2}(x,y) = \lambda e^{-\lambda x}\lambda e^{-\lambda y} \qquad x, y > 0$. Then

$$f_{M,V}(m,v) = \frac{\lambda^2 e^{-\lambda(m+\sqrt{2v})} e^{-\lambda(m-\sqrt{2v})}}{\sqrt{v/2}} \qquad \underbrace{\underbrace{m - \sqrt{2v} > 0}_{m > \sqrt{2v} > 0}}_{\frac{m^2}{2} > v > 0}$$

$$= \frac{\lambda^2 e^{-2\lambda m}}{\sqrt{v/2}} \qquad 0 < v < \frac{m^2}{2}$$

As a check, find $f_M(m)$:

$$f_M(m) = \int_0^{m^2/2} \frac{\lambda^2 e^{-2\lambda m}}{\sqrt{v/2}} dv = \lambda^2 e^{-2\lambda m} 2m$$

This is an Erlang RV for the sum of 2 exponential RV's

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6.23 (a)

$$f_{X_1,X_2}(x,y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}$$

$$f_{MV}(m,v) = \frac{1}{2\pi\sqrt{v}/2} e^{-\frac{1}{2}\left[(m+\sqrt{2v})^2 + (m-\sqrt{2v})^2\right]}$$

$$= \frac{e^{-\frac{1}{2}[2m^2+2v]}}{2\pi\sqrt{v}/2} = \frac{e^{-m^2}}{\sqrt{2\pi}\frac{1}{2}} \cdot \frac{e^{-2v}}{\sqrt{4\pi v}}$$

Gaussian chi-square

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6.24

~~4.55~~ a) Let the auxiliary function be $W = Y$ then

$$J(z,w) = \begin{vmatrix} \frac{w}{(1-z)^2} & \frac{z}{1-z} \\ 0 & 1 \end{vmatrix} = \frac{w}{(1-z)^2}$$

and

$$f_{Z,W}(z,w) = f_{XY}\left(\frac{zw}{1-z}, w\right)\frac{|w|}{(1-z)^2}$$

so

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}\left(\frac{zw}{1-z}, w\right)\frac{|w|}{(1-z)^2}dw$$

b)
$$\begin{aligned} f_Z(z) &= \int_0^{\infty} \alpha e^{-\alpha zw/(1-z)}\alpha e^{-\alpha w}\frac{w}{(1-z)^2}dw \qquad 0 \le z \le 1 \\ &= \frac{\alpha^2}{(1-z)^2}\int_0^{\infty} we^{-\frac{\alpha}{(1-z)}w} = \frac{\alpha^2}{(1-z)^2}\frac{(1-z)^2}{\alpha^2} \\ &= 1 \end{aligned}$$

That is Z is unif. dist. in [0,1].

(c)

$$f_Z(z) = \int_{x_m}^{\infty} \alpha^2 \frac{x_m^{2\alpha}(1-z)^{\alpha+1}}{(zw)^{\alpha+1}w^{\alpha+1}}\frac{w}{(1-z)^2}dw$$

$$= \frac{\alpha^2 x_m^{2\alpha}(1-z)^{\alpha+1}}{z^{\alpha+1}(1-z)^2}\int_{x_m}^{\infty}\frac{1}{w^{2\alpha+1}}dw$$

$$-\frac{1}{2}w^{-2\alpha}\Big|_{x_m}^{\infty}$$

$$= \frac{\alpha^2(1-z)^{\alpha-1}}{z^{\alpha+1}}\frac{x_m^{2\alpha}}{2x_m^{2\alpha}} = \frac{\alpha^2}{2}\frac{(1-z)^{\alpha-1}}{z^{\alpha+1}}$$

$0 < z < 1$

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6.25

$$Z = X/Y \qquad J(x,y) = \begin{vmatrix} 1/y & -x/y^2 \\ 0 & 1 \end{vmatrix} = \frac{1}{y} = \frac{1}{w}$$
$$W = Y$$

$$f_{ZW}(z,w) = f_{XY}(zw, w)\,|w|$$

$$f_Z(z) = \int_0^\infty \alpha e^{-\alpha z w}\, \alpha e^{-\alpha w}\, w\, dw$$

$$= \alpha \int_0^\infty w\, e^{-(\alpha z+1)w}\, dw$$

$$= \frac{\alpha}{\alpha z+1} \int_0^\infty w\,(\alpha z+1)\, e^{-(\alpha z+1)w}\, dw$$

(the integral is the mean of an exponential RV)

$$= \frac{\alpha}{(\alpha z+1)^2} \qquad z > 0$$

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6.26

~~4.56~~ $U = X^2$ Four $(\sqrt{u}, \sqrt[4]{v})$ $(+\sqrt{u}, -\sqrt[4]{v})$
$V = Y^4$ Roots: $(-\sqrt{u}, \sqrt[4]{v})$ $(-\sqrt{u}, -\sqrt[4]{v})$

$$J_{XY} = \begin{vmatrix} 2x & 0 \\ 0 & 4y^3 \end{vmatrix} = 8|xy^3|$$

$$\begin{aligned} f_{UV}(u,v) &= \sum_i \frac{f_{XY}(x_i, y_i)}{8|x_i y_i^3|} \qquad u > 0, v > 0 \\ &= \frac{1}{8\sqrt{u}^4\sqrt{v^3}} \left[\frac{2e^{-(u-2\rho\sqrt{u}^4\sqrt{v}+\sqrt{v})/2(1-\rho^2)}}{2\pi\sqrt{1-\rho^2}} \right. \\ &\quad \left. + \frac{2e^{-(u+2\rho\sqrt{u}^4\sqrt{v}+\sqrt{v})/2(1-\rho^2)}}{2\pi\sqrt{1-\rho^2}} \right] \end{aligned}$$

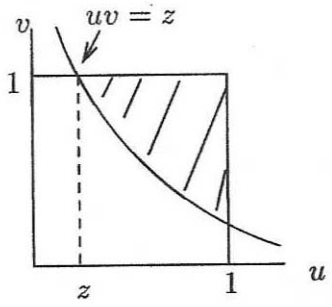
6.27

~~4.57~~ Defining two auxiliary functions $U = X_2$, $V = X_3$

$$J_{X_1X_2X_3} = \begin{vmatrix} x_2x_3 & x_1x_3 & x_1x_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = |x_2x_3|$$

$$f_{Z_1,U,V}(z,u,v) = \frac{f_{\underline{X}}(z/uv, u, v)}{uv} = \frac{1}{uv} \qquad \begin{matrix} 0 < \frac{z}{uv} < 1 \\ 0 < u < 1 \\ 0 < v < 1 \end{matrix}$$

$$= \frac{1}{uv} \qquad \begin{matrix} 0 < z < uv \\ 0 < u < 1 \\ 0 < v < 1 \end{matrix}$$



$$\begin{aligned} f_Z(z) &= \int\int_{\text{shaded area}} \frac{dudv}{uv} = \int_z^1 \int_{z/u}^1 \frac{1}{uv} dvdu \\ &= \frac{1}{2}(\ln z)^2 \end{aligned}$$

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6.28

~~1.58~~ Use spherical coordinates:

(a)

$$X = R\cos\Theta\sin\phi \qquad Y = R\sin\Theta\sin\phi \qquad Z = R\cos\phi$$

$$J(r,\theta,\phi) = \begin{vmatrix} \cos\theta\sin\phi & \sin\theta\sin\phi & \cos\phi \\ -r\sin\theta\sin\phi & r\cos\theta\sin\phi & 0 \\ r\cos\theta\cos\phi & r\sin\theta\cos\phi & -r\sin\phi \end{vmatrix} = |-r^2\sin\phi|$$

$$\begin{aligned} f_{R,\Theta,\phi}(r,\theta,\phi) &= f_{\underline{X}}(r\cos\theta\sin\phi, r\sin\theta\sin\phi, r\cos\phi)r^2\sin\phi \\ &= \frac{e^{-r^2/2}}{\sqrt{2\pi}^3}r^2\sin\phi \end{aligned}$$

$$\begin{aligned} f_R(r) &= \int_0^{2\pi} d\theta \int_0^{\pi} d\phi \frac{e^{-r^2/2}}{\sqrt{2\pi}^3}r^2\sin\phi \\ &= \int_0^{\pi} \frac{r^2 e^{-r^2/2}}{\sqrt{2\pi}}\sin\phi d\phi \\ &= \sqrt{\frac{2}{\pi}}r^2 e^{-r^2/2} \qquad r > 0 \end{aligned}$$

(b) Let $W = R^2$ then

$$f_W(w) = \frac{f_R(\sqrt{w})}{2\sqrt{w}} = \sqrt{\frac{2}{\pi}}\frac{w e^{-w/2}}{2\sqrt{w}}$$

$$= \frac{\sqrt{w}\, e^{-w/2}}{\sqrt{2\pi}}$$

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6.29

$$Y_1 = X_1 \qquad X_1 = Y_1$$

$$Y_2 = X_1 + X_2 \qquad X_2 = -Y_1 + Y_2$$

$$Y_3 = X_2 + X_3 \qquad X_3 = Y_1 - Y_2 + Y_3$$

$$Y_4 = X_3 + X_4 \qquad X_4 = -Y_1 + Y_2 - Y_3 + Y_4$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \qquad |A| = 1.$$

(a) $$f_{\underline{Y}}(\underline{y}) = f_{\underline{X}}(y_1, -y_1+y_2, y_1-y_2+y_3, -y_1+y_2-y_3+y_4)$$

(b) $$f_{\underline{Y}}(\underline{y}) = \frac{1}{(2\pi)^2} \exp\left\{-\frac{1}{2}\left[y_1^2 + (-y_1+y_2)^2 + (y_1-y_2+y_3)^2 + (-y_1+y_2-y_3+y_4)^2 \right]\right\}$$

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6.3 Expected Values of Vector Random Variables

6.30

$$\begin{aligned}\mathcal{E}[M] &= \frac{1}{2}\mathcal{E}[X_1] + \frac{1}{2}\mathcal{E}[X_2] \\ \mathcal{E}[V] &= \mathcal{E}\left[\frac{(X_1 - M)^2}{2} + \frac{(X_2 - M)^2}{2}\right] = \frac{1}{8}\mathcal{E}[(X_1 - X_2)^2] \\ &= \frac{1}{8}[\mathcal{E}[X_1^2] - 2\mathcal{E}[X_1 x_2] + \mathcal{E}[X_2^2]].\end{aligned}$$

In 6.30c M and V are independent, so $E[MV] = E[M]E[V] = 0$.

6.31

a) $$\mathcal{E}[Z] = \int_0^1 z\frac{1}{2}(\ln z)^2 dz = \frac{1}{4}\left[z^2(\ln z)^2 - z^2 \ln z + \frac{z^2}{2}\right]_0^1 = \frac{1}{8}$$

b)
$$\begin{aligned}\mathcal{E}[X_1X_2X_3] &= \int_0^1\int_0^1\int_0^1 x_1x_2x_3 dx_1dx_2dx_3 \\ &= \int_0^1 x_1dx_1 \int_0^1 x_2dx_2 \int_0^1 x_3dx_3 = \left(\frac{1}{2}\right)^3\end{aligned}$$

6.32

$$m_X = E[\vec{X}] = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}$$

$$VAR[X_K] = VAR[5 + N_K] = 1$$

$$K_K = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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6.33

$$E[X(t)] = E[A\sin t] = E[A]\sin t = \tfrac{1}{2}\sin t$$

$$E[X(t_1)X(t_2)] = E[A\sin t_1 \, A\sin t_2] \qquad E[A^2] = \int_0^1 x^2\,dx = \frac{1}{3}$$

$$= E[A^2]\sin t_1 \sin t_2$$

$$= \tfrac{1}{3}\sin t_1 \sin t_2$$

$$COV(X(t_1), X(t_2)) = \tfrac{1}{3}\sin t_1 \sin t_2 - \tfrac{1}{4}\sin t_1 \sin t_2$$

$$= \frac{1}{12}\sin t_1 \sin t_2.$$

$$\underline{m}_X = \begin{bmatrix} \frac{1}{2}\sin t_1 \\ \frac{1}{2}\sin t_2 \\ \frac{1}{2}\sin t_3 \end{bmatrix}$$

$$\underline{K}_X = \frac{1}{12}\begin{bmatrix} \sin^2 t_1 & \sin t_1 \sin t_2 & \sin t_1 \sin t_3 \\ \sin t_1 \sin t_2 & \sin^2 t_2 & \sin t_2 \sin t_3 \\ \sin t_1 \sin t_3 & \sin t_2 \sin t_3 & \sin^2 t_3 \end{bmatrix}$$

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6.34

(a)

$$E[X] = 3\cdot\tfrac{1}{2} = \tfrac{3}{2}$$
$$E[Y] = \tfrac{1}{8}$$
$$E[Z] = \tfrac{7}{8}$$

$$\underline{m}_X = \begin{bmatrix} \frac{3}{2} \\ \frac{1}{8} \\ \frac{7}{8} \end{bmatrix}$$

$$E[XY] = 3\cdot 1\cdot\tfrac{1}{8} = \tfrac{3}{8}$$
$$E[XZ] = 1\cdot 1\cdot\tfrac{3}{8} + 2\cdot 1\cdot\tfrac{3}{8} + 3\cdot 1\cdot\tfrac{1}{8} = \tfrac{12}{8} = \tfrac{3}{2}$$
$$E[YZ] = 1\cdot 1\cdot\tfrac{1}{8} = \tfrac{1}{8}$$
$$E[X^2] = 3\cdot\tfrac{1}{2}\cdot\tfrac{1}{2} + \left(3\cdot\tfrac{1}{2}\right)^2 = \tfrac{3}{4} + \tfrac{9}{4} = 3$$
$$E[Y^2] = \tfrac{1}{8}$$
$$E[Z^2] = \tfrac{7}{8}$$

$$\underline{K}_X = \begin{bmatrix} \frac{3}{2} & \frac{3}{8} - \frac{3}{2}\frac{1}{8} & \frac{3}{2} - \frac{3}{2}\frac{7}{8} \\ - & \frac{1}{8} - \left(\frac{1}{8}\right)^2 & \frac{1}{8} - \frac{1}{8}\frac{7}{8} \\ - & - & \frac{7}{8} - \left(\frac{7}{8}\right)^2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{2} & \frac{3}{16} & \frac{3}{16} \\ \frac{3}{16} & \frac{7}{64} & \frac{1}{64} \\ \frac{3}{16} & \frac{1}{64} & \frac{7}{64} \end{bmatrix}$$

(b)

$$E[X] = 1 \quad E[Y] = 0 \quad E[Z] = 1$$
$$E[X^2] = 1 \quad E[Y^2] = 0 \quad E[Z^2] = 1$$

$$\underline{m}_X = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\underline{K}_X = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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6.35

Note symmetry in x, y, and z so number of calculations can be reduced drastically.

$$E[X] = \frac{2}{3}\int_0^1 x(x+1)\,dx = \frac{2}{3}\left[\frac{x^3}{3} + \frac{x^2}{2}\right]_0^1 = \frac{5}{9}$$

$$= E[Y] = E[Z]$$

$$E[X^2] = \frac{2}{3}\int_0^1 x^2(x+1)\,dx = \frac{2}{3}\left[\frac{x^4}{4} + \frac{x^3}{3}\right]_0^1 = \frac{7}{18}$$

$$VAR[X] = \frac{7}{18} - \left(\frac{5}{9}\right)^2 = \frac{13}{162}$$

$$E[XY] = \frac{2}{3}\int_0^1\int_0^1 xy\left(x + y + \frac{1}{2}\right)dx\,dy$$

$$= \frac{2}{3}\int_0^1 x\left(\int_0^1 \left(yx + y^2 + \frac{1}{2}y\right)dy\right)$$

$$\frac{1}{2}x + \frac{1}{3} + \frac{1}{4}$$

$$= \frac{2}{3}\int_0^1 \left(\frac{1}{2}x^2 + \frac{7}{12}x\right)dx$$

$$= \frac{2}{3}\left[\frac{1}{2}\cdot\frac{1}{3} + \frac{7}{12}\cdot\frac{1}{2}\right] = \frac{11}{36}$$

$$E[XY] - E[X]E[Y] = \frac{11}{36} - \left(\frac{5}{9}\right)^2 = -\frac{1}{81}$$

$$\underline{m}_X = \begin{bmatrix} \frac{5}{9} \\ \frac{5}{9} \\ \frac{5}{9} \end{bmatrix} \qquad \underline{K}_X = \begin{bmatrix} \frac{13}{162} & -\frac{2}{162} & -\frac{2}{162} \\ -\frac{2}{162} & \frac{13}{162} & -\frac{2}{162} \\ -\frac{2}{162} & -\frac{2}{162} & \frac{13}{162} \end{bmatrix}$$

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6.36

$$E[X] = \int_{-1}^{1} x\,\frac{3}{4}(1-x^2)\,dx = \frac{3x^2}{8} - \frac{3x^4}{16}\Big|_{-1}^{1} = 0$$

$$E[Y] = 0$$

$$E[Z] = 0$$

$$E[X^2] = \int_{-1}^{1} x^2\,\frac{3}{4}(1-x^2)\,dx = \frac{3}{4}\cdot\frac{x^3}{3} - \frac{3}{4}\cdot\frac{x^5}{5}\Big|_{-1}^{1} = \frac{1}{5}$$

$$E[Y^2] = \frac{1}{5}$$

$$E[Z^2] = \frac{1}{5}$$

$$E[XY] = \int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} xy\,\frac{3}{2\pi}\sqrt{1-x^2-y^2}\,dx\,dy$$

$$= \int_{-1}^{1}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{3}{2\pi}\,y\,\sqrt{1-y^2}\sin u\,\sqrt{1-y^2}\cos u\,\sqrt{1-y^2}\cos u\,du\,dy$$

$$= \int_{-1}^{1} \frac{3}{2\pi}\,y\,(1-y^2)^{3/2}\left(-\frac{\cos^3 u}{3}\right)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} dy = 0$$

$$m = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad K = \begin{bmatrix} 1/5 & 0 & 0 \\ 0 & 1/5 & 0 \\ 0 & 0 & 1/5 \end{bmatrix}$$

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6.37

(a) X_2 is Binomial with parameters 3 and $\frac{p}{3}$

$$E[X_2] = 3 \cdot \frac{p}{3} = p$$

(b)
$$E[X_1 X_2] = \sum_{i=0}^{3} \sum_{j=0}^{3-i} ij \frac{3!}{i!\,j!\,(3-i-j)!} \left(\frac{p}{3}\right)^i \left(\frac{p}{3}\right)^j \left(1-2\frac{p}{3}\right)^{3-i-j}$$

$$= \sum_{i=0}^{3} i \frac{3!}{i!} \left(\frac{p}{3}\right)^i \sum_{j=1}^{3-i} \frac{\left(\frac{p}{3}\right)^{j-1}}{(j-1)!} \frac{\left(1-2\frac{p}{3}\right)^{3-i-j}}{(3-i-j)!}$$

$$\sum_{j'=0}^{3-i-1} \frac{\left(\frac{p}{3}\right)^{j'}}{j'!} \frac{\left(1-2\frac{p}{3}\right)^{3-i-j'-1}}{(3-i-j'-1)!}$$

$$\frac{1}{(3-i-1)!} \sum_{j'=0}^{3-i-1} \binom{3-i-1}{j'} \left(\frac{p}{3}\right)^{j'} \left(1-2\frac{p}{3}\right)^{3-i-j'-1}$$

$$\left(\frac{p}{3} + 1 - 2\frac{p}{3}\right)^{3-i-1}$$

$$E[X_1 X_2] = \sum_{i=1}^{3} \frac{3!}{(i-1)!} \left(\frac{p}{3}\right)^i \left(\frac{p}{3}\right) \frac{\left(1-\frac{p}{3}\right)^{3-i-1}}{(3-i-1)!}$$

$$= 3!\left(\frac{p}{3}\right) \sum_{i'=0}^{2} \frac{\left(\frac{p}{3}\right)^{i'+1}}{i'!} \frac{\left(1-\frac{p}{3}\right)^{3-i'-2}}{(3-i'-2)!}$$

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6.38

$$E[E[N_1N_2|T]] = E[(\lambda_1 T)(\lambda_2 T)]$$
$$= \lambda_1\lambda_2 E[T^2]$$

$$\text{COV}(N_1,N_2) = \lambda_1\lambda_2 E[T^2] - \lambda_1 E[T]\lambda_2 E[T]$$
$$= \lambda_1\lambda_2\left[\frac{2}{\alpha^2}\right] - \lambda_1\lambda_2\frac{1}{\alpha^2}$$
$$= \lambda_1\lambda_2\frac{1}{\alpha^2}$$

$$E[E[N_1^2|T]] = E[\lambda_1 T + (\lambda_1 T)^2]$$
$$= \lambda_1 E[T] + \lambda_1^2 E[T^2]$$
$$= \frac{\lambda_1}{\alpha} + \lambda_1^2\frac{2}{\alpha^2}$$

$$\text{VAR}[N_1] = \frac{\lambda_1}{\alpha} + \lambda_1^2\frac{2}{\alpha^2} - \left(\frac{\lambda_1}{\alpha}\right)^2$$
$$= \frac{\lambda_1^2}{\alpha^2} + \frac{\lambda_1}{\alpha}$$

$$\underline{m}_N = \begin{bmatrix} \lambda_1/\alpha \\ \lambda_2/\alpha \\ \lambda_3/\alpha \end{bmatrix} \qquad \underline{K}_N = \begin{bmatrix} \frac{\lambda_1^2}{\alpha^2}+\frac{\lambda_1}{\alpha} & \frac{\lambda_1\lambda_2}{\alpha^2} & \frac{\lambda_1\lambda_3}{\alpha^2} \\ \frac{\lambda_2\lambda_1}{\alpha^2} & \frac{\lambda_2^2}{\alpha^2}+\frac{\lambda_2}{\alpha} & \frac{\lambda_2\lambda_3}{\alpha^2} \\ \frac{\lambda_1\lambda_3}{\alpha^2} & \frac{\lambda_2\lambda_3}{\alpha^2} & \frac{\lambda_3^2}{\alpha^2}+\frac{\lambda_3}{\alpha} \end{bmatrix}$$

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6.39

X_i iid Gaussian

$\underline{m}_X = \underline{0}$ $K_X = I$

$U = X_1$

$V = X_1 + X_2$

$W = X_1 + X_2 + X_3$

(a) $\underline{Y} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \underline{X}$ $\underline{m}_Y = A\underline{m}_X = \underline{0}$

$$K_Y = AK_XA^T = AA^T$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

(b) $K_{XY} = K_XA^T = A^T = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

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6.40

a) & b)

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \qquad m_Y = \begin{bmatrix} E[X_1] \\ E[X_1]+E[X_2] \\ E[X_2]+E[X_3] \\ E[X_3]+E[X_4] \end{bmatrix}$$

$$K_{XY} = K_X A^T = \begin{bmatrix} k_{11}+k_{21} & k_{11}+k_{12} & k_{12}+k_{13} & k_{13}+k_{14} \\ k_{21} & k_{21}+k_{22} & k_{22}+k_{23} & k_{23}+k_{24} \\ k_{31} & k_{31}+k_{32} & k_{32}+k_{33} & k_{33}+k_{34} \\ k_{41} & k_{41}+k_{42} & k_{42}+k_{43} & k_{43}+k_{44} \end{bmatrix}$$

$$K_Y = \begin{bmatrix} k_{11} & k_{11}+k_{12} & k_{12}+k_{13} & k_{13}+k_{14} \\ k_{11}+k_{21} & k_{11}+k_{12}+k_{21}+k_{22} & k_{12}+k_{13}+k_{22}+k_{23} & k_{13}+k_{14}+k_{23}+k_{24} \\ k_{21}+k_{31} & k_{31}+k_{32}+k_{32}+k_{33} & k_{22}+k_{23}+k_{32}+k_{33} & k_{23}+k_{24}+k_{33}+k_{34} \\ k_{31}+k_{41} & k_{31}+k_{32}+k_{41}+k_{42} & k_{32}+k_{33}+k_{42}+k_{43} & k_{33}+k_{34}+k_{43}+k_{44} \end{bmatrix}$$

$k_{11} = VAR[X_1]$ $k_{22} = VAR(X_2)$ $k_{33} = VAR(X_3)$ $k_{44} = VAR(X_4)$

$k_{12} = k_{21} = COV(X_1, X_2)$ $k_{13} = k_{31} = COV(X_1, X_3)$ ...

c) $k_{12} = k_{21} = k_{13} = k_{31} = \ldots = 0$

d) $$m_Y = \begin{bmatrix} E[X_1] \\ E[X_1]+E[X_2] \\ \vdots \\ E[X_{n-1}]+E[X_n] \end{bmatrix} \qquad K_Y = \begin{bmatrix} k_{11} & k_{11}+k_{12} & & 0 \\ k_{11}+k_{12} & k_{11}+k_{12}+k_{21}+k_{22} & & 0 \\ 0 & & \ddots & \vdots \\ \vdots & 0 & 0 \cdots & 2(k_{n-1,n-1}+k_{n-1,n})+1 \end{bmatrix}$$

$k_{nn}+k_{n-1,n-1}+2k_{n-1,n}$

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6.41

a) $$m_Y = Am_X = \begin{bmatrix} m+\frac{1}{2}m+\frac{1}{4}m+\frac{1}{8}m \\ m+\frac{1}{2}m+\frac{1}{4}m \\ m+\frac{1}{2}m \\ m \end{bmatrix} = \begin{bmatrix} 15/8\,m \\ 7/4\,m \\ 3/2\,m \\ m \end{bmatrix}$$

$$K_{XY} = K_X A^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ 1/4 & 1/2 & 1 & 0 \\ 1/8 & 1/4 & 1/2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ 1/4 & 1/2 & 1 & 0 \\ 1/8 & 1/4 & 1/2 & 1 \end{bmatrix}$$

$$K_Y = AK_{XY} = \begin{bmatrix} 1 & 1/2 & 1/4 & 1/8 \\ 0 & 1 & 1/2 & 1/4 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ 1/4 & 1/2 & 1 & 0 \\ 1/8 & 1/4 & 1/2 & 1 \end{bmatrix}$$

$$K_Y = \begin{bmatrix} 1.32812 & 0.65625 & 0.3125 & 0.125 \\ 0.65625 & 1.3125 & 0.625 & 0.25 \\ 0.3125 & 0.625 & 1.25 & 0.5 \\ 0.125 & 0.25 & 0.5 & 1 \end{bmatrix}$$

b) $$m_Y = Am_X = \begin{bmatrix} 4m \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$K_{XY} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

$$K_Y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

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6.42

4.70 a)
$$\begin{aligned}\phi_V(w) &= E[e^{jwV}] = E[e^{jw(zX+bY+c)}] \\ &= e^{jwc}\phi_{X,Y}(aw, bw)\end{aligned}$$

b)
$$\begin{aligned}\phi_V(w) &= e^{jwc}\phi_{X,Y}(aw, bw) \\ &= e^{jwc}\phi_U(w_1 + 2w_1)\phi_V(w_1 + w_2)|_{w_1=aw, w_2=bw} \\ &= e^{jwc}\phi_U((a+2b)w)\phi_V((a+b)w) \\ &= e^{jwc}\exp\left[-\frac{1}{2}(aw+2bw)^2\right]\exp\left[-\frac{1}{2}(aw+bw)^2\right] \\ &= e^{jwc}\exp\left[-\frac{1}{2}w^2(a^2+4ab+4b^2+a^2+2ab+b^2)\right] \\ &= e^{jwc}\exp\left[-\frac{1}{2}w^2(2a^2+6ab+5b^2)\right] \\ f_V(v) &= \frac{1}{\sqrt{2\pi(2a^2+6ab+5b^2)}}\exp\left[-\frac{(v-c)^2}{2(a^2+6ab+5b^2)}\right]\end{aligned}$$

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6.43

4.71 a)

$$\begin{aligned}
\phi_{X,Y}(w_1,w_2) &= E[e^{jw_1X+jw_1Y}] \\
&= E\left[\exp\left[jw_1\left(\frac{1}{\sqrt{2}}V-\frac{1}{\sqrt{2}}W\right)+jw_2\left(\frac{1}{\sqrt{2}}V+\frac{1}{\sqrt{2}}W\right)\right]\right] \\
&= E\left[\exp\left[\left(j\frac{1}{\sqrt{2}}w_1+j\frac{1}{\sqrt{2}}w_2\right)V+\left(-j\frac{1}{\sqrt{2}}w_1+j\frac{1}{\sqrt{2}}w_2\right)W\right]\right] \\
&= \phi_V\left(\frac{1}{\sqrt{2}}w_1+\frac{1}{\sqrt{2}}w_2\right)\phi_W\left(-\frac{1}{\sqrt{2}}w_1+\frac{1}{\sqrt{2}}w_2\right) \\
&= \exp\left[-\frac{1}{2}(1+\rho\left(\frac{1}{\sqrt{2}}w_1+\frac{1}{\sqrt{2}}\right)^2\right] \\
&\quad \exp\left[-\frac{1}{2}(1-\rho)\left(-\frac{1}{\sqrt{2}}w_1+\frac{1}{\sqrt{2}}w_2\right)^2\right] \\
&= \exp\left[-\frac{1}{4}(1+\rho(w_1+w_2)^2-\frac{1}{4}(1-\rho(-w_1+w_2)^2\right] \\
&= \exp\left[-\frac{1}{2}(w_1^2+2w_1w_2\rho+w_2^2)\right]
\end{aligned}$$

b)

$$\begin{aligned}
\frac{\partial\phi_{X,Y}(w_1,w_2)}{\partial w_1^2\partial w_2} &= \frac{\partial}{\partial w_1^2}\left[(-w_1\rho-w_2)\exp\left(-\frac{1}{2}(w_1^2+2w_1w_2\rho+w_2^2)\right)\right] \\
&= \frac{\partial}{\partial w_1}\left\{-\rho\exp\left[-\frac{1}{2}(w_1^2+w_1w_2\rho+w_2^2)\right]\right. \\
&\quad \left.+(w_1\rho+w_2)(w_1+w_2\rho)\cdot\exp\left[-\frac{1}{2}(w_1^2+2w_1w_2\rho+w_2^2)\right]\right\} \\
&= p(w_1+w_2\rho)\exp\left[-\frac{1}{2}(w_1^2+2w_1w_2\rho)+w_2^2)\right] \\
&\quad +p(w_1+w_2\rho)\exp\left[-\frac{1}{2}(w_1^2+w_1w_2\rho+w_2^2)\right] \\
&\quad +(w_1\rho+w_1)\exp\left[-\frac{1}{2}(w_1^2+2w_1w_2\rho)+w_2^2\right] \\
&\quad -(w_1\rho+w_2)(w_1+w_2\rho)^2\exp\left[-\frac{1}{2}(w_1^2+w_1w_2\rho+w_2^2)\right]
\end{aligned}$$

$$E[X^2Y] = \frac{1}{j^3}\frac{\partial\phi_{X,Y}(w_1,w_2)}{\partial w_1^2\partial w_2}\bigg|_{w_1=0,w_2=0} = 0$$

c)

$$\begin{aligned}
\phi_{X',Y'}(w_1,w_2) &= E[e^{jw_1X'+jw_2Y'}] \\
&= E[e^{jw_1(X+a)+jw_2(Y+b)}] \\
&= e^{jw_1a+jw_2b}\phi_{X,Y}(w_1,w_2) \\
&= \exp[jw_1a+jw_2b]\exp\left[-\frac{1}{2}(w_1^2+2w_1w_2\rho+w_2^2)\right]
\end{aligned}$$

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6.44

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} U \\ V \end{pmatrix}$$

$$\begin{pmatrix} U \\ V \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \begin{pmatrix} X \\ Y \end{pmatrix} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$$

a) $$\begin{aligned} \phi_{X,Y}(w_1, w_2) &= E[e^{jw_1X + jw_2Y}] \\ &= E[\exp[jw_1(aU+bV) + jw_2(cU+dV)]] \\ &= E[\exp[(jw_1a + jw_2c)U + (jw_1b + jw_2d)V]] \\ &= \phi_{U,V}(aw_1 + cw_2, bw_1 + dw_2) \end{aligned}$$

b) $$\begin{aligned} \frac{\partial^2}{\partial w_1 \partial w_2}\phi_{X,Y}(w_1, w_2) &= \frac{\partial}{\partial w_1}\left[\frac{\partial \phi}{\partial u}c + \frac{\partial \phi}{\partial v}d\right] \\ &= c\left[\frac{\partial^2 \phi}{\partial u^2}a + \frac{\partial^2 \phi}{\partial u \partial v}b\right] + d\left[\frac{\partial \phi}{\partial u \partial v}a + \frac{\partial \phi}{\partial v^2}b\right] \end{aligned}$$

$$\begin{aligned} E[XY] &= \frac{\partial^2}{\partial w_1 \partial w_2}\phi_{X,Y}(w_1, w_2)\Big|_{w_1=w_2=0} \\ &= acE[U^2] + (bc+ad)E[UV] + bdE[V^2] \,. \end{aligned}$$

We check this result by direct calculation:

$$\begin{aligned} E[XY] &= E[(aU+bV)(cU+dV)] \\ &= acE[U^2] + (bc+ad)E[UV] + bdE[V^2] \end{aligned}$$

6.45

4.73 a) Poisson RV X_1 and X_2 with rates σ_1 and σ_2

$$G_1(z) = e^{+\sigma_1(z-1)}, \qquad G_2(z) = e^{+\sigma_2(z-1)}$$

X_1 and X_2 are independent

$$\begin{aligned} G_{X_1,X_2}(z_1, z_2) &= E[z_1^{X_1} z_2^{X_2}] = E[z_1^{X_1}]E[z_2^{X_2}] \\ &= G_1(z_1)G_2(z_2) \\ &= e^{+\sigma_1(z_1-1)+\sigma_2(z_2-1)} \end{aligned}$$

b) $G_1(z) = (q+ps)^n$, $G_2(z) = (q+pz)^m$

$$G_{X_1,X_2}(z_1, z_2) = G_1(z_1)G_2(z_2) = (q+pz_1)^n(q+pz_1)^m$$

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6.46

4.74 a)
$$\begin{aligned} G_X(z) &= G_{X,Y}(z_1,z_2)|_{z_1=z,z_2=1} = e^{(\sigma_1+\beta)(z-1)} \\ G_Y(z) &= e^{(\sigma_2+\beta)(z-1)} \end{aligned}$$

$\therefore X$ and Y are Poisson RVs.

b) $G_Z(z) = E[z^{X+Y}] = G_{X,Y}(z_1,z_2)|_{z_1=z_2=z} = e^{(\sigma_2+\sigma_2)(z_1)+\beta(z^2-1)}$
Z is not a Poisson RV.

6.47

4.75 a)
$$\begin{aligned} G_{X,Y}(z_1,z_2) &= E[z_1^X z_2^Y] = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} z_1^j z_2^k P[X=j,Y=k] \qquad j+k \le n \\ &= \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} z_1^j z_2^k \frac{n!}{j!k!(n-j-k)!} p_1^j p_2^k (1-p_1-p_2)^{n-j-k} \\ &= \sum_{j=0}^{n} \frac{z_1^j n! p_1^j}{j!(n-j)!} \sum_{k=0}^{n-j} \frac{(n-j)! z_2^k}{k!(n-j-k)!} p_2^k (1-p_1-p_2)^{n-j-k} \\ &= \sum_{j=0}^{n} \frac{n!}{j!(n-j)!} (p_1 z_1)^j (p_2 z_2 + 1 - p_1 - p_2)^{n-j} \\ &= (p_1 z_1 + p_2 z_2 + 1 - p_1 - p_2)^n \end{aligned}$$

b)
$$\begin{aligned} E[XY] &= \left.\frac{\partial G(z_1,z_2)}{\partial z_1 \partial z_2}\right|_{z_1=z_2=1} \\ &= \left.\frac{\partial}{\partial z_1} n p_2 (p_1 z_1 + p_2 z_2 + 1 - p_1 - p_2)^{n-1}\right|_{z_1=z_2=1} \\ &= p_1 p_2 n(n-1) \end{aligned}$$

$$\begin{aligned} E[X] &= \left.\frac{\partial G(z_1,1)}{\partial z_1}\right|_{z_1=1} \\ &= \left.\frac{\partial}{\partial z_1}(p_1 z_1 + 1 - p_1)^n\right|_{z_1=1} \\ &= np_1 \end{aligned}$$

$$E[Y] = np_2$$

$$\begin{aligned} COV[X,Y] &= E[XY] - E[X]E[Y] \\ &= p_1 p_2 n^2 - p_1 p_2 n - n^2 p_1 p_2 \\ &= -n p_1 p_2 \end{aligned}$$

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6.48

$$G_{XY}(z_1,z_2) = e^{\alpha_1(z_1-1)+\alpha_2(z_2-1)+\beta(z_1z_2-1)}$$

$$\frac{\partial}{\partial z_2} G_{XY}(z_1,z_2) = e^{(\,)}\left[\alpha_1+\beta z_2\right]$$

$$\frac{\partial^2}{\partial z_2 \partial z_1} G_{XY}(z_1,z_2)\Big| = e^{(\,)}[\alpha_1+\beta z_2][\alpha_2+\beta z_1] + e^{(\,)}(\beta)$$

$$z_1=z_2=1$$

$$= (\alpha_1+\beta)(\alpha_2+\beta)+\beta$$

$$\frac{\partial^2}{\partial z_2 \partial z_1} G_{XY}(z_1,z_2) = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} j z_1^{j-1} k z_2^{k-1} P[X=j,Y=k]$$

$$= E[XY]$$

$$\Rightarrow E[XY] = (\alpha_1+\beta)(\alpha_2+\beta)+\beta$$

$$Cov(X,Y) = \alpha_1\alpha_2+\alpha_1\beta+\alpha_2\beta+\beta-\alpha_1\alpha_2$$

from 6.46 we know X and Y are Poisson RV's

$$E[X]=\alpha_1 \quad VAR[X]=\alpha_1$$

$$E[Y]=\alpha_2 \quad VAR[Y]=\alpha_2$$

$$\underline{m}_X = \begin{bmatrix}\alpha_1\\ \alpha_2\end{bmatrix} \quad \underline{K}_X = \begin{bmatrix}\alpha_1 & \beta(\alpha_1+\alpha_2+1)\\ \beta(\alpha_1+\alpha_2+1) & \alpha_2\end{bmatrix}$$

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6.49 from 6.47

$E[X] = np_1$ $\quad COV(X,Y) = -np_1p_2$

$E[Y] = np_2$ $\quad VAR(X) = np_1(1-p_1)$

$VAR(Y) = np_2(1-p_2)$

$$\underline{m}_X = \begin{bmatrix} np_1 \\ np_2 \end{bmatrix} \qquad \underline{K}_X = \begin{bmatrix} np_1(1-p_1) & -np_1p_2 \\ -np_1p_2 & np_2(1-p_2) \end{bmatrix}$$

6.50

a) $$\det\begin{bmatrix} 1-\lambda & 1/4 \\ 1/4 & 1-\lambda \end{bmatrix} = (1-\lambda)^2 - 1/16 = \lambda^2 - 2\lambda + 15/16 = \left(\lambda - \frac{5}{4}\right)\left(\lambda - \frac{3}{4}\right)$$

$\lambda_1 = \frac{5}{4}$

$\lambda_2 = \frac{3}{4}$

$$\begin{bmatrix} 1 & 1/4 \\ 1/4 & 1 \end{bmatrix}\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \frac{5}{4}\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1/4 \\ 1/4 & 1 \end{bmatrix}\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \frac{3}{4}\begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

$e_1 + \frac{1}{4}e_2 = \frac{5}{4}e_1$ $\qquad e_1 + \frac{1}{4}e_2 = \frac{3}{4}e_1$

$-\frac{1}{4}e_1 + \frac{1}{4}e_2 = 0$ $\qquad \frac{1}{4}e_1 + \frac{1}{4}e_2 = 0$

$e_1 = [1, 1]^T$ - form $\qquad e_2 = [1, -1]^T$ - form

$e_1 = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]^T$ $\qquad e_2 = \left[\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right]$

b) $$P = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \qquad P^TP = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$P^T K_k P = \left(\frac{1}{\sqrt{2}}\right)^2 \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1/4 \\ 1/4 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5/4 & 0 \\ 0 & 3/4 \end{bmatrix}$$

c) $$\underline{X} = P\underline{Y} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} y_1 + y_2 \\ y_1 - y_2 \end{bmatrix}$$

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6.51

a) $\det K_X = -\lambda(\lambda - \frac{3}{2})^2$

$\lambda_1 = 0$ $\quad e_1 = \frac{1}{\sqrt{3}}[1 \ 1 \ 1]^T$

$\lambda_2 = \frac{3}{2}$ $\quad e_2 = \frac{1}{\sqrt{6}}[1 \ -1 \ 2]^T$

$\lambda_3 = \frac{3}{2}$ $\quad e_3 = \frac{1}{\sqrt{2}}[1 \ -1 \ 0]^T$

b) $P = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \end{bmatrix}$ $\qquad P^TP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$P^T K_X P = P^T \begin{bmatrix} 0 & -3/2\sqrt{6} & 3/2\sqrt{2} \\ 0 & -3/2\sqrt{6} & -3/2\sqrt{2} \\ 0 & 3/\sqrt{6} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3/2 & 0 \\ 0 & 0 & 3/2 \end{bmatrix}$$

c) $$X = PY = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}}y_1 + \frac{1}{\sqrt{6}}y_2 + \frac{1}{\sqrt{2}}y_3 \\ \frac{1}{\sqrt{3}}y_1 - \frac{1}{\sqrt{6}}y_2 - \frac{1}{\sqrt{2}}y_3 \\ \frac{1}{\sqrt{3}}y_1 + \frac{2}{\sqrt{6}}y_2 \end{bmatrix}$$

6.52

$$K_X = E[(\bar{X} - \bar{m}_X)(\bar{X} - \bar{m}_X)^T]$$

$$\begin{aligned} \bar{a}^T K_X \bar{a} &= a^T E[(\bar{X} - \bar{m}_X)(\bar{X} - \bar{m}_X)^T] a \\ &= E[\bar{a}^T(\bar{X} - \bar{m}_X)(\bar{X} - \bar{m}_X)^T \bar{a}] \\ &= E[(\bar{a}^T(X - \bar{m}_X))^2] \geq 0 \end{aligned}$$

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6.53

$\underline{b} = P^T \underline{a}$ where $K_X = P \Lambda P^T$

(a)

$$\underline{b}^T K_X \underline{b} = (P^T \underline{a})^T K_X (P\underline{a}) = \underline{a}^T P K_X P \underline{a}$$

$$= \underline{a}^T \Lambda \underline{a}$$

$$= \sum_{i=0}^{n} a_i^2 \lambda_i \geq 0 \quad \text{if } \lambda_i > 0 \text{ for all } i \text{ and } a_i > 0 \text{ for some } i$$

$\Rightarrow K_X$ is positive definite

(b) If K_X is positive definite, then

$$a^T K_X \underline{a} > 0 \quad \text{for any nonzero vector } \underline{a}$$

in particular if $\underline{a}$ is an eigenvector, then

$$0 < a^T K_X \underline{a} = a^T \lambda_i \underline{a} = \lambda_i \underbrace{a^T a}_{>0}$$

$$\Rightarrow \lambda_i > 0$$

$\therefore$ all evalues of K are positive.

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6.4 Jointly Gaussian Random Vectors

6.54

a) $$f_{\bar{X}}(\bar{x}) = \frac{e^{-\frac{1}{2}(\bar{x}-\bar{m}_x)^T K_x^{-1}(\bar{x}-\bar{m}_x)}}{2\pi\sqrt{2}}$$

$$\det[K] = 2$$

$$K_x^{-1} = \begin{bmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{bmatrix} \qquad \bar{m}_x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

b) $$\begin{bmatrix} x_1-1 & x_2 \end{bmatrix}\begin{bmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{bmatrix}\begin{bmatrix} x_1-1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1-1 & x_2 \end{bmatrix}\begin{bmatrix} 3(x_1-1)+x_2 \\ (x_1-1)+3x_2 \end{bmatrix}\left(\frac{1}{4}\right)$$

$$= \frac{1}{4}\left[3(x_1-1)^2 + x_2(x_1-1) + x_2(x_1-1) + 3x_2^2\right]$$

$$= \frac{3}{4}\left[(x_1-1)^2 + x_2^2 + \frac{2}{3}x_2(x_1-1)\right]$$

$$f_{\bar{X}}(\bar{x}) = \frac{e^{-\frac{1}{2}\left(\frac{3}{4}\right)\left[(x_1-1)^2 + x_2^2 + \frac{2}{3}x_2(x_1-1)\right]}}{2\pi\sqrt{2}}$$

c) $$f_{X_1}(x_1) = \frac{e^{-\frac{1}{2}(x_1-1)^2/(3/2)}}{\sqrt{2\pi}\sqrt{3/2}}$$

$$f_{X_2}(x_2) = \frac{e^{-\frac{1}{2}(x_2)^2/(3/2)}}{\sqrt{2\pi}\sqrt{3/2}}$$

d) $$\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \qquad P = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} \qquad \text{for } K_X$$

$$A = P^T = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} \qquad K_Y = \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \qquad m_Y = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

e) $$\det[K_Y] = 2 \qquad K_Y^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$f_Y(y) = \frac{e^{-\frac{1}{2}\left[(y_1+\frac{1}{\sqrt{2}})^2 + (y_2+\frac{1}{\sqrt{2}})^2/2\right]}}{2\pi\sqrt{2}}$$

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6.55

a) $\det[K_X] = \frac{3}{2}$

$$K^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -\frac{1}{3} & 0 & 1 \end{bmatrix} \qquad m_X = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$f_{\bar{X}}(\bar{x}) = \frac{e^{-\frac{1}{2}(\bar{x}-\bar{m})^T K^{-1}(\bar{x}-\bar{m})}}{(2\pi)^{3/2}\sqrt{3/2}}$$

b) $$K^{-1}(\bar{x}-\bar{m}) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -\frac{1}{3} & 0 & 1 \end{bmatrix}\begin{bmatrix} x_1-1 \\ x_2 \\ x_3-2 \end{bmatrix} = \begin{bmatrix} x_1-1-(x_3-2) \\ x_2 \\ -\frac{1}{3}(x_1-1)+(x_3-2) \end{bmatrix}$$

$$(\bar{x}-\bar{m})^T K^{-1}(\bar{x}-\bar{m}) = \begin{bmatrix} x_1-1 & x_2 & x_3-2 \end{bmatrix}\begin{bmatrix} x_1-1-(x_3-2) \\ x_2 \\ -\frac{1}{3}(x_1-1)+(x_3-2) \end{bmatrix}$$

$$= (x_1-1)^2 - (x_1-1)(x_3-2) + x_2^2 - \tfrac{1}{3}(x_1-1)(x_3-2) + (x_3-2)^2$$

$$f_{\bar{X}}(x) = \frac{e^{-\frac{1}{2}[(x_1-1)^2 + x_2^2 + (x_3-2)^2 - 4/3(x_1-1)(x_3-2)]}}{(2\pi)^{3/2}\sqrt{3/2}}$$

c) $$f_{X_1}(x_1) = \frac{e^{-\frac{1}{2}(x_1-1)^2/(3/2)}}{\sqrt{2\pi}\sqrt{3/2}} \qquad f_{X_2}(x_2) = \frac{e^{-\frac{1}{2}x_2^2}}{\sqrt{2\pi}} \qquad f_{X_3}(x_3) = \frac{e^{-\frac{1}{2}(x_3-2)^2/(3/2)}}{\sqrt{2\pi}\sqrt{3/2}}$$

d) $$P = \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -1 & 0 & 0 \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \qquad \Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$K_Y = \Lambda$
$A = P^T$

e) $\det(K_Y) = 2$ $\qquad m_Y = [0 \quad -1/\sqrt{2} \quad 3/\sqrt{2}]^T$

$$f_Y(\bar{y}) = \frac{e^{-\frac{1}{2}[y_1^2 + (y_2+1/\sqrt{2})^2 + (y_3-3/\sqrt{2})^2/2]}}{(2\pi)^{3/2}\sqrt{2}}$$

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6.56

a) $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ $K_U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $m_U = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$K_Y = AK_UA^T = AA^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$

b) $\det(K_Y) = 1$

$K_Y^{-1} = \begin{bmatrix} +2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

$u^T K_Y^{-1} u^T = 2x^2 + 2y^2 + z^2 - 2xy - 2yz$

$f_Y(\vec{y}) = \dfrac{e^{-\frac{1}{2}[2x^2+2y^2+z^2-2xy-2yz]}}{(2\pi)^{3/2}}$

c) $f(y,z|x) = \dfrac{f(x,y,z)}{f_X(x)} = \dfrac{e^{-\frac{1}{2}[x^2+2y^2+z^2-2xy-2yz]}}{2\pi}$

$f_X(x) = \dfrac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$

d) $f(z|x,y) = \dfrac{f(x,y,z)}{f(x,y)} = \dfrac{e^{-\frac{1}{2}[y^2+z^2-2yz]}}{\sqrt{2\pi}}$

$f(x,y) = \dfrac{e^{-\frac{1}{2}[2x^2+y^2-2xy]}}{2\pi}$ $K' = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ $(K')^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$

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6.57

a) $K_X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$K_Y = AA^T = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ $K_Y(y_1, y_2, y_3) = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

b) $\det(K_Y) = 4$

$K_Y^{-1} = \frac{1}{4}\begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{bmatrix}$

$$f_Y(y_1, y_2, y_3) = \frac{e^{-\frac{1}{2}\left(\frac{1}{4}\right)\left[3x_1^2 + 4x_2^2 + 3x_3^2 - 4x_1x_2 - 4x_2x_3 + 2x_1x_3\right]}}{(2\pi)^{3/2}(2)}$$

c) $$f(y_1, y_2) = \frac{e^{-\frac{1}{2}\left[2y_1^2 + 2y_2^2 - 2y_1y_2\right]\left(\frac{1}{3}\right)}}{2\pi\sqrt{3}}$$

$K' = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ $\det(K') = 3$ $K'^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

$$f(y_1, y_3) = \frac{e^{-\frac{1}{2}\left(\frac{1}{2}\right)\left[x^2 + z^2\right)}}{2\pi(2)}$$

$K' = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ $\det(K') = 4$

d) $P = \begin{bmatrix} 1/2 & -1/\sqrt{2} & 1/2 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/2 & 1/\sqrt{2} & 1/2 \end{bmatrix}$ $\Lambda = \begin{bmatrix} 2-\sqrt{2} & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2+\sqrt{2} \end{bmatrix}$ $\Lambda = K_Z$ $A = P^T$

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6.58

$\underline{Y} = AR\underline{b} + \underline{N}$ $\quad A = \text{diag}[a_k]$ $\quad a_k > 0$

$R = R^T$

$\underline{N}$ Gaussian $\quad m_{\underline{N}} = \underline{0} \quad K_{\underline{N}} = I$

(a)

$$m_{\underline{Y}} = E[\underline{Y}] = E[AR\underline{b} + \underline{N}] = AR\underline{b} + E[\underline{N}] = AR\underline{b}$$

$$K_{\underline{Y}} = E[(\underline{Y} - m_{\underline{Y}})(\underline{Y} - m_{\underline{Y}})^T] = E[\underline{N}\underline{N}^T] = I$$

$$f_{\underline{Y}}(\underline{y}) = \frac{\exp\{-\frac{1}{2}(\underline{y} - AR\underline{b})^T(\underline{y} - AR\underline{b})\}}{(2\pi)^{k/2}}$$

(b)

$$\underline{Z} = (AR)^{-1}\underline{Y} = (AR)^{-1}((AR)\underline{b} + \underline{N})$$

$$= \underline{b} + (AR)^{-1}\underline{N}$$

$$m_{\underline{Z}} = E[\underline{Z}] = \underline{b} + (AR)^{-1}\underbrace{E[\underline{N}]}_{\underline{0}} = \underline{b}$$

$$K_{\underline{Z}} = E[(\underline{Z} - m_{\underline{Z}})(\underline{Z} - m_{\underline{Z}})^T]$$

$$= E[(AR)^{-1}\underline{N}((AR)^{-1}\underline{N})^T]$$

$$= E[(AR)^{-1}\underline{N}\underline{N}^T(AR)^{-1T}]$$

$$= (AR)^{-1}\underbrace{E[\underline{N}\underline{N}^T]}_{I}(AR)^{-1T}$$

$$= R^{-1}A^{-1}(R^{-1}A^{-1})^T = (R^{-1}A^{-1})(A^{-1T})(R^{-1T}) = R^{-1}A^{-2}R^{-1}$$

(diagonal, symmetric)

$$K_{\underline{Z}}^{-1} = (R^{-1}A^{-2}R^{-1})^{-1} = RA^2R$$

$$f_{\underline{Z}}(\underline{z}) = \frac{\exp\{-\frac{1}{2}(\underline{z} - \underline{b})RA^2R(\underline{z} - \underline{b})^T\}}{(2\pi)^{k/2}\,|RA^2R|}$$

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6.59

a) $K_2 = \begin{bmatrix} 3/2 & 0 \\ 0 & 1 \end{bmatrix}$ $Q_2 = \begin{bmatrix} 2/3 & 0 \\ 0 & 1 \end{bmatrix}$ $\det(K_2) = \frac{3}{2}$

$K_3 = \begin{bmatrix} 3/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 3/2 \end{bmatrix}$ $Q_3 = \begin{bmatrix} 3/4 & 0 & -1/4 \\ 0 & 1 & 0 \\ -1/4 & 0 & 3/4 \end{bmatrix}$ $\det(K_3) = 2$

b) $f(x_3 | x_1, x_2) = \dfrac{f(x_1, x_2, x_3)}{f(x_1, x_2)}$

$$f(x_1, x_2, x_3) = \frac{e^{-\frac{1}{2}(\frac{1}{4})\left[3x_1'^2 + 4x_2^2 + 3x_3'^2 - 2(x_1')(x_3')\right]}}{\sqrt{2}\,(2\pi)^{3/2}}$$

$x_1' = (x_1 - 1)$

$x_3' = (x_3 - 2)$

$$f(x_1, x_2) = \frac{e^{-\frac{1}{2}\left[\frac{2}{3}x_1'^2 + x_2'^2\right]}}{(2\pi)\sqrt{3/2}}$$

$$f(x_3 | x_1, x_2) = \frac{e^{-\frac{1}{2}\left[\frac{1}{12}x_1'^2 + \frac{3}{4}x_3'^2 - \frac{1}{2}x_1'x_3'\right]}}{\sqrt{2\pi}\,\sqrt{4/3}}$$

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6.60

If we write out the quadratic form in the exponent we obtain:

$$\sum_{j=1}^{n}\sum_{k=1}^{n} Q_{jk}(x_j - m_j)(x_k - m_k) - \sum_{j=1}^{n-1}\sum_{k=1}^{n-1} Q_{jk}(x_j - m_j)(x_k - m_k)$$

$$= Q_{nn}(x_n - m_n)^2 + 2(x_n - m_n)\sum_{j=1}^{n-1} Q_{jk}(x_j - m_j)$$

$$= Q_{nn}\left\{(x_n - m_n)^2 + 2(x_n - m_n)\frac{1}{Q_{nn}}\sum_{j=1}^{n-1} Q_{jk}(x_j - m_j)\right\}$$

$$= Q_{nn}\left\{(x_n - m_n)^2 + 2(x_n - m_n)B + B^2\right\} - Q_{nn}B^2$$

$$= Q_{nn}\left\{(x_n - m_n) + B\right\}^2 - Q_{nn}B^2$$

where $B = \dfrac{1}{Q_{nn}}\sum_{j=1}^{n-1} Q_{jk}(x_j - m_j)$

In the first line, all the terms in the second summation are contained in the first summation. We then completed the square to obtain an expression involving x_k, its mean $m_k - B$, and its variance $1/Q_{kk}$. The term $Q_{kk}B^2$ is part of the normalization constant. We therefore conclude that:

$$f_{X_n}(x_n \mid x_1, ..., x_{n-1}) = \frac{\exp\left\{-\dfrac{1}{2Q_{nn}}\left(x - m_n + \dfrac{1}{Q_{nn}}\sum_{j=1}^{n-1} Q_{jk}(x_j - m_j)\right)^2\right\}}{\sqrt{2\pi Q_{nn}}}$$

We see that the conditional mean of x_n is a linear function of the observations x_1, x_2,…, x_{n-1}.

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6.61

a) $A=\begin{bmatrix}1&1&1\\0&1&0\\0&0&1\end{bmatrix}$ $m_Z=\begin{bmatrix}3\\1\\2\end{bmatrix}$ $K_Z=AK_XA^T=\begin{bmatrix}5&1&3/2\\1&1&0\\2&0&3/2\end{bmatrix}$

$$f_Z(z)=\frac{e^{-\frac{1}{2}(z-3)^2/5}}{\sqrt{2\pi}\sqrt{5}}$$

b) $A=\begin{bmatrix}1&1&1\\0&1&0\\0&0&1\end{bmatrix}$ $m_Z=\begin{bmatrix}0\\0\\0\end{bmatrix}$ $K_Z=AK_YA^T=\begin{bmatrix}14&5&6\\5&2&2\\6&2&3\end{bmatrix}$

$$f_Z(z)=\frac{e^{-\frac{1}{2}z^2/14}}{\sqrt{2\pi}\sqrt{14}}$$

c) $A=\begin{bmatrix}1&1&1\\0&1&0\\0&0&1\end{bmatrix}$ $m_Z=\begin{bmatrix}0\\0\\0\end{bmatrix}$ $\mathrm{VAR}[Z]=10=\sum_i\sum_j \mathrm{COV}(Y_i,Y_j)$

$$f_Z(z)=\frac{e^{-\frac{1}{2}z^2/10}}{\sqrt{2\pi}\sqrt{10}}$$

6.62

$$\Phi_X(\omega)=e^{j\omega^T m-\frac{1}{2}\omega^T K\omega}$$

$$j[\omega_1\ \omega_2]\begin{bmatrix}1\\0\end{bmatrix}=j\omega_1$$

$$-\frac{1}{2}[\omega_1\ \omega_2]\begin{bmatrix}3/2&-1/2\\-1/2&3/2\end{bmatrix}\begin{bmatrix}\omega_1\\\omega_2\end{bmatrix}=-\frac{1}{2}[\omega_1\ \omega_2]\begin{bmatrix}3/2\omega_1-1/2\omega_2\\-1/2\omega_1+3/2\omega_2\end{bmatrix}$$

$$=-\frac{1}{2}\left[\frac{3}{2}\omega_1^2-\omega_1\omega_2+\frac{3}{2}\omega_2^2\right]$$

$$\Phi_X(\omega)=e^{j\omega_1-\frac{1}{4}[3\omega_1^2+3\omega_2^2-2\omega_1\omega_2]}$$

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6.63

a) $$[\omega_1\ \omega_2\ \omega_3]\begin{bmatrix} 1 & -1/2 & -1/2 \\ -1/2 & 1 & -1/2 \\ -1/2 & -1/2 & 1 \end{bmatrix}\begin{bmatrix}\omega_1 \\ \omega_2 \\ \omega_3\end{bmatrix}$$

$$= \omega_1^2+\omega_2^2+\omega_3^2 - \omega_1\omega_2 - \omega_2\omega_3 - \omega_1\omega_3$$

$$\Phi_X(\omega) = e^{-\frac{1}{2}[\omega_1^2+\omega_2^2+\omega_3^2-\omega_1\omega_2-\omega_1\omega_3-\omega_2\omega_3]}$$

b) No

6.64

$$\begin{aligned}
\phi_{X,Y}(w_1,w_2) &= E[e^{jw_1X}e^{jw_2Y}] \\
&= E[E[e^{jw_1X}e^{jw_2Y}|Y]] \\
&= E[e^{jw_2}E[e^{jw_1X}|Y]] \\
&= E[e^{jw_2Y}e^{jw_1[m_1+\rho\frac{\sigma_1}{\sigma_2}(Y-m_2)]-\frac{1}{2}w_2^2\sigma_1^2(1-\rho^2)}] \\
&= \exp(jw_1m_1 - jw_1\rho\frac{\sigma_1}{\sigma_2}m_2 - \frac{1}{2}w_1^2\sigma_1^2(1-\rho^2))E[e^{jw_2Y+jw_1\rho\frac{\sigma_1}{\sigma_2}Y}] \\
&= \exp(jw_1m_2 - jw_1\rho\frac{\sigma_1}{\sigma_2}m_2 - \frac{1}{2}w_1^2\sigma_1^2(1-\rho^2)) \\
&\quad \cdot\exp\left(j\left(w_2+w_1\rho\frac{\sigma_1}{\sigma_2}\right)m_2 - \frac{1}{2}\sigma_2^2\left(w_2+w_1\rho\frac{\sigma_1}{\sigma_2}\right)^2\right) \\
&= \exp(jw_1m_1 + jw_2m_2 - \frac{1}{2}w_1^2\sigma_1^2 - w_1w_2\rho\sigma_1\sigma_2 - \frac{1}{2}w_2^2\sigma_2^2)
\end{aligned}$$

6.65

$$\begin{aligned}
\phi_{X_1,\ldots,X_n}(w_1,\ldots,w_n) &= E[e^{j\mathbf{w}^TX}] \qquad \mathbf{w}=(w_1,w_2,\ldots,w_n)^T \\
&= \int \exp(j\mathbf{w}^TX)\frac{1}{(2\pi)^{n/2}|K^{1/2}}\exp\{-\frac{1}{2}(X-m)^TK^{-1}(X-m)\}dX \\
&= \frac{1}{(2\pi)^{n/2}|K|^{1/2}}\int \exp\{-\frac{1}{2}[X^TK^{-1}X - m^TK^{-1}X - jw^TX - X^TK^{-1}m \\
&\quad -X^TK^{-1}Kjw + (m+jkw)^TK^{-1}(m+jKw) \\
&\quad -(m+jKw)^TK^{-1}(m+jKw) + m^TK^{-1}m\}dX \\
&\qquad (w^TK = X^Tw = w_1x_1 + w_2x_2 + \ldots + w_nx_n) \\
&= \frac{1}{(2\pi)^{n/1}|K|^{1/2}}\int \exp\{-\frac{1}{2}(x-m-jw)^TK^{-1}(x-m-jw) \\
&\quad -jw^TK^TK^{-1}m - jm^TK^{-1}Kw + w^TK^TK^{-1}Kw\}dX \\
&= \exp[jw^Tm - \frac{1}{2}w^TKw] \\
&\qquad (w^Tm = m^Tw, \quad K = K^T)
\end{aligned}$$

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6.66

$$\phi_{X,Y}(w_1,w_2) = \exp\left(-\frac{1}{2}w_1^2\sigma_1^2 - \frac{1}{2}w_1^2\sigma_2^2 - w_1w_2\sigma_1\sigma_2\right)$$

$$E[X^2Y^2] = \frac{\partial^4}{\partial w_1^2\partial w_2^2}\phi_{X,Y}(w_1,w_2)\bigg|_{w_1=w_2=0}$$

$$\frac{\partial\phi}{\partial w_1} = \phi\cdot(-\sigma_1^2w_1 - w_2\sigma_1\sigma_2p)$$

$$\frac{\partial^2\phi}{\partial w_1^2} = \phi(-\sigma_1^2w_1 - w_2\sigma_1\sigma_2p)^2 + \phi(-\sigma_1^2)$$

$$\frac{\partial^3\phi}{\partial w_1^2\partial w_2} = \phi(-\sigma_1^2w_1 - w_2\sigma_1\sigma_2p)^2(-\sigma_2^2w_2 - w_1\sigma_1\sigma_2p)$$
$$+\phi\cdot 2(-\sigma_1^2w_1 - w_2\sigma_1\sigma_2p)(-\sigma_1\sigma_2p)$$
$$+\phi(-\sigma_2^2w_2 - w_1\sigma_1\sigma_2p)(-\sigma_1^2)$$

$$\frac{\partial^4\phi}{\partial w_1^2\partial w_2^2} = \phi(-\sigma_1^2w_1 - w_2\sigma_1\sigma_2p)^3(-\sigma_2^2w_2 - w_1\sigma_1\sigma_2p)$$
$$+\phi\cdot 2(-\sigma_1^2w_1 - w_2\sigma_1\sigma_2p)(-\sigma_1\sigma_2p)(-\sigma_2^2w_2 - w_1\sigma_1\sigma_2p)$$
$$+\phi(-\sigma_1^2w_1 - w_2\sigma_1\sigma_2p)(-\sigma_2^2)$$
$$+\phi\cdot(-\sigma_2^2w_2 - w_1\sigma_1\sigma_2p)\cdot 2(-\sigma_1^2w_1 - w_2\sigma_1\sigma_2p)(-\sigma_1\sigma_2p)$$
$$+\phi\cdot 2(-\sigma_1\sigma_2p)(-\sigma_1\sigma_2p)$$
$$+\phi(-\sigma_2^2w_2 - w_1\sigma_1\sigma_2p)^2(-\sigma_1^2) + \phi(-\sigma_2^2)(-\sigma_1^2)$$

$$E[X^2Y^2] = \frac{\partial^4\phi(w_1,w_2)}{\partial w_1^2\partial w_2^2)}\bigg|_{w_1=w_2=0} = 2(\sigma_1\sigma_2p)^2 + \sigma_1^2\sigma_2^2$$
$$= E[X^2]E[Y^2] + 2E^2[XY]$$

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6.67

4.88 The joint characteristic function for (X_1, X_2, X_3, X_4) is:

$$\Phi_{\underline{X}}(\underline{w}) = e^{-\frac{1}{2}\underline{w}^T K \underline{w}}$$

where

$$\underline{w}^T K \underline{w} = (w_1, w_2, w_3, w_4)\,[E[X_i X_j]] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix} = \sum_{i=1}^{4}\sum_{j=1}^{4} E[X_i X_j] w_i w_j$$

Expanding the exponential in a power series:

$$\Phi_{\underline{X}}(\underline{w}) = 1 - \frac{1}{2}\underline{w}^T K \underline{w} + \frac{1}{8}(\underline{w}^T K \underline{w})^2 + \ldots$$

From the moment theorem we know that $E[X_1 X_2 X_3 X_4]$ is the coefficient of $w_1 w_2 w_3 w_4$ in the above series. This coefficient appears in the third term:

$$\begin{aligned} \frac{1}{8}(\underline{w}^T K \underline{w})^2 &= \frac{1}{8}\left(\sum_{ij} E[X_i X_j] x_i x_j\right)\left(\sum_{i'j'} E[X_{i'} X_{j'}] w_{i'} w_{j'}\right) \\ &= \frac{1}{8}\sum_{ij}\sum_{i'j'} E[X_i X_j] E[X_{i'} X_{j'}] w_i w_j w_{i'} w_{j'} \end{aligned}$$

By grouping all terms that give $w_1 w_2 w_3 w_4$ we find

$$\begin{aligned} E[X_1 X_2 X_3 X_4] &= \frac{1}{8}[8E[X_1 X_2]E[X_3 X_4] + 8E[X_1 X_3]E[X_2 X_4] + 8E[X_1 X_4]E[X_2 X_3]] \\ &= E[X_1 X_2]E[X_3 X_4] + E[X_1 X_3]E[X_2 X_4] + E[X_1 X_4]E[X_2 X_3]\ . \end{aligned}$$

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6.5 Estimation of Random Variables

6.68

(i)

X \ Y	-1	0	1
-1	1/6	1/6	
0			1/3
1	1/6	1/6	

(ii)

X \ Y	-1	0	1
-1	1/9	1/9	1/9
0	1/9	1/9	1/9
1	1/9	1/9	1/9

(iii)

X \ Y	-1	0	1
-1	1/3		
0		1/3	
1			1/3

From Problem 5.61:

(i) $\text{COV}(X,Y) = 0$

$\text{VAR}(X) = (-1)^2\frac{1}{3} + (1)^2\frac{1}{3} = \frac{2}{3}$

$\text{VAR}(Y) = \frac{2}{3}$

$\rho_{XY} = 0$

(ii) $\text{COV}(X,Y) = 0$

$\text{VAR}(X) = \frac{2}{3}$

$\text{VAR}(Y) = \frac{2}{3}$

$\rho_{XY} = 0$

(iii) $\text{COV}(X,Y) = \frac{2}{3}$

$\text{VAR}(X) = \frac{2}{3}$

$\text{VAR}(Y) = \frac{2}{3}$

$\rho_{XY} = \dfrac{\frac{2}{3}}{\sqrt{\frac{2}{3}}\sqrt{\frac{2}{3}}} = 1.$

(a) $\hat{Y} = \rho_{XY}\dfrac{\sigma_Y}{\sigma_X}(X - m_X) + m_X$

$= 0$

(i) $\hat{Y} = 0$

(iii) $\hat{Y} = X$

(b) $\hat{Y}(-1) = E[Y|-1] = -\frac{1}{2}$

$\hat{Y}(0) = E[Y|0] = 1$

$\hat{Y}(+1) = E[Y|+1] = -\frac{1}{2}$

(i) $\hat{Y}(-1) = 0$

$\hat{Y}(0) = 0$

$\hat{Y}(+1) = 0$

(iii) $\hat{Y}(-1) = -1$

$\hat{Y}(0) = 0$

$\hat{Y}(+1) = +1$

(c) $P_Y(j|X=k) = \dfrac{P[X=k|Y=j]P[Y=j]}{P[X=k]} = P_X(k|Y=j]$

Since $P[Y=j] = 1/3 = P[X=k]$

$\Rightarrow$ ML and MAP estimates the same

(i) $\hat{Y}(-1) = 0$ or $+1$

$\hat{Y}(0) = 1$

$\hat{Y}(+1) = 0$ or 1

(ii) $\hat{Y}(-1) = 0, 1,$ or -1

$\hat{Y}(0) =$ "

$\hat{Y}(+1) =$ "

(iii) $\hat{Y}(-1) = -1$

$\hat{Y}(0) = 0$

$\hat{Y}(+1) = +1$

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6.68

(d) Mean Square Error

	(i)	(ii)	(iii)
Lin MSE	$(-1)^2\frac{1}{6} + (1)^2\frac{1}{6} = \frac{1}{3}$	$\left((-1)^2\cdot\frac{1}{9} + (1)^2\frac{1}{9}\right)2 = \frac{4}{9}$	0
MAMSE	$\left(\frac{1}{2}\right)^2\frac{1}{6}\cdot 2\cdot 2 = \frac{1}{6}$	$\frac{4}{9}$	0
ML/MAP	$\frac{1}{3}$	$\frac{4}{9}$	0

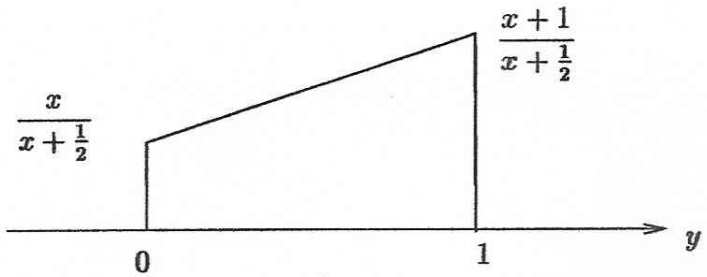
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6.69

~~4.90~~ a) $\hat{Y} = -\frac{1}{11}\left(x - \frac{7}{12}\right) + \frac{7}{12}$

$$\mathcal{E}[(Y-\hat{Y})^2] = VAR[Y](1-\rho^2) = \frac{11}{144}\left(1 - \left(\frac{1}{11}\right)^2\right) = \frac{55}{726} = .075$$

b)



$$f(y|x) = \frac{x+y}{x+\frac{1}{2}} \qquad 0 \le y \le 1$$

$$\max \; y = 1 \Rightarrow \hat{Y} = 1$$

$$\therefore \; \mathcal{E}[(Y-\hat{Y})^2] = \mathcal{E}[(Y-1)^2] = \mathcal{E}[Y^2] - 2\mathcal{E}[Y] + 1 = 0.25$$

c) $\hat{Y} = \mathcal{E}[Y|x] = \int_0^1 \frac{x+y}{x+\frac{1}{2}} y\,dy = \frac{\frac{x}{2}+\frac{1}{3}}{x+\frac{1}{2}}$

$$\mathcal{E}[(Y-\hat{Y})^2] = \int_0^1 dx \int_0^1 \left(y - \frac{\frac{x}{2}+\frac{1}{3}}{x+\frac{1}{2}}\right)^2 (x+y)\,dy$$

$$= \int_0^1 dx \int_0^1 \left[y^2 - 2\frac{\frac{x}{2}+\frac{1}{3}}{x+\frac{1}{2}} + \left(\frac{\frac{x}{2}+\frac{1}{3}}{x+\frac{1}{2}}\right)^2\right](x+y)\,dy$$

$$\mathcal{E}[(Y-\hat{Y})^2] = \int_0^1 \left[\left(\frac{x}{3}+\frac{1}{4}\right) - \frac{\frac{2}{2}+\frac{1}{3}}{x+\frac{1}{2}}\left(x+\frac{2}{3}\right) + \left(\frac{\frac{x}{2}+\frac{1}{3}}{x+\frac{1}{2}}\right)^2\left(x+\frac{1}{2}\right)\right] dx$$

$$= \int_0^1 \left[\left(\frac{x}{3}+\frac{1}{4}\right) + \frac{-\left(\frac{x}{2}+\frac{1}{3}\right)\left(x+\frac{2}{3}\right) + \left(\frac{x}{2}+\frac{1}{3}\right)^2}{x+\frac{1}{2}}\right] dx$$

$$= \int_0^1 \left(\left(\frac{x}{3}+\frac{1}{4}\right) - \frac{\left(\frac{x}{2}+\frac{1}{3}\right)^2}{x+\frac{1}{2}}\right] dx$$

$$= \int_0^1 \left(\frac{x}{3}+\frac{1}{4}-\frac{x}{4}-\frac{5}{24}-\frac{\frac{1}{144}}{x+\frac{1}{2}}\right) dx$$

$$= \int_0^1 \left(\frac{x}{12}+\frac{1}{24}-\frac{1}{144}\frac{1}{x+\frac{1}{2}}\right) dx$$

$$= \frac{1}{24}+\frac{1}{24}-\frac{1}{144}\ln\left(x+\frac{1}{2}\right)\Big|_0^1$$

$$= \frac{1}{12}-\frac{1}{144}\ln 3$$

$$= 0.07570$$

This is slightly better than the linear predictor.

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6.70

$X_i = s + N_i \quad i = 1, 2, 3$ N_i indep, zero mean unit variance Gaussian RV's

$$f_{\underline{X}}(\underline{x}) = \frac{1}{(\sqrt{2\pi})^3} e^{-\frac{1}{2}\sum_{i=1}^{3}(x_i - s)^2}$$

To find maximum with respect to s, minimize the argument in the exponent:

$$0 = \frac{d}{ds}\sum_{i=1}^{3}(x_i - s)^2 = \sum_{i=1}^{3} 2(x_i - s)$$

$$\sum_{i=1}^{3} x_i = 3s$$

$$s = \frac{1}{3}\sum_{i=1}^{3} x_i$$

This is the sample mean of the three received signals.

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6.71 From Problem 5.63

$E[N_1] = np \qquad E[N_2] = 2np$

$VAR[N_1] = npq \qquad VAR[N_2] = 2npq$

$COV(N_1, N_2) = npq \qquad \rho_{N_1,N_2} = \dfrac{npq}{\sqrt{npq}\sqrt{2npq}} = \dfrac{1}{\sqrt{2}}$

(a) $\hat{N}_2 = \rho_{N_1,N_2}\dfrac{\sigma_{N_2}}{\sigma_{N_1}}(N_1 - m_1) + m_2 = \dfrac{1}{\sqrt{2}}\dfrac{\sqrt{2npq}}{\sqrt{npq}}(N_1 - np) + 2np$

$\hat{N}_2 = N_1 + np$ ← expected value of additional arrivals

(b) $\hat{N}_2 = E[N_2 | N_1] = E[(N_2 - N_1) + N_1 | N_1]$

$= np + N_1$ same as (a)

(c) $P[N_2 = j+k \mid N_1 = j] = \dfrac{\binom{n}{k}p^k(1-p)^{n-k}\binom{n}{j}p^j(1-p)^{n-j}}{\binom{n}{j}p^j(1-p)^{n-j}}$

$= \binom{n}{k}p^k(1-p)^{n-k} = P[N_2 = k]$ ← max at $k \approx np$

$\hat{N}_2 = np + N_1$

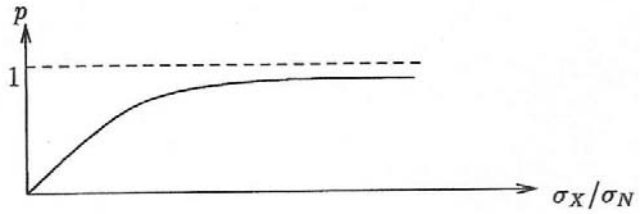
(d) $\hat{N}_1 = \dfrac{1}{\sqrt{2}}\dfrac{\sqrt{npq}}{\sqrt{2npq}}(N_2 - 2np) + np = \dfrac{1}{2}N_2$ Linear Est

$P_{N_1}(j \mid N_2 = k) = \dfrac{\binom{n}{k-j}p^{k-j}(1-p)^{n-k+j}\binom{n}{j}p^j(1-p)^{n-j}}{\binom{2n}{k}p^k(1-p)^{2n-k}}$

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6.72 a) $$COV[XY] = E[XY] = E[X(X+N)] = VAR[X] = \sigma_X^2$$

$$\rho = \frac{COV[XY]}{\sigma_X \sigma_Y} = \frac{\sigma_X^2}{\sigma_X(\sigma_X^2 + \sigma_N^2)^{1/2}} = \left(\frac{1}{1 + \sigma_N^2/\sigma_X^2}\right)^{1/2}$$



b)
$$\begin{aligned}
\hat{X} &= \rho\frac{\sigma_X}{\sigma_Y}Y = \rho^2 Y \\
&= \frac{COV[XY]}{\sigma_Y^2}Y \\
&= \frac{\sigma_X^2}{\sigma_Y^2}Y \\
MSE &= E[(X-\hat{X})^2] \\
&= E[(X-\rho^2 Y)^2] \\
&= VAR[X^2] + \rho^4 VAR[Y] - 2\rho^2 COV[X,Y] \\
&= \sigma_X^2 + \rho^2\sigma_X^2 - 2\rho^2\sigma_X^2 \\
&= \sigma_X^2(1-\rho^2)
\end{aligned}$$

— continued —

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(6.72) From Example 6.26

the MAP estimator is the same as the MMSE estimator

On the other hand, the ML receiver is given by

$$\hat{X}_{ML} = \frac{\sigma_X}{\rho\sigma_Y}(Y - m_Y) + m_X = \frac{\sigma_X}{\rho\sigma_Y} Y$$

$$= \frac{\sigma_X\sqrt{1+\sigma_N^2/\sigma_X^2}}{\sqrt{\sigma_X^2+\sigma_N^2}} Y = Y$$

Thus the ML estimator gives a different estimate.

The MSE for the ML estimator is

$$MSE_{ML} = E[(X - \hat{X}_{ML})^2]$$

$$= E[(X-Y)^2]$$

$$= E[N^2]$$

$$= \sigma_N^2$$

In comparison to the MAP estimator MSE we have

$$MSE_{MAP} = \sigma_X^2(1-\rho^2) = \sigma_X^2\left(1 - \frac{1}{1+\frac{\sigma_N^2}{\sigma_X^2}}\right) = \sigma_N^2 \underbrace{\frac{\sigma_X^2}{\sigma_X^2+\sigma_N^2}}_{<1}$$

$$\therefore \quad MSE_{MAP} < MSE_{ML}$$

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6.73

$$f(x,y,z) = \frac{2}{3}(x+y+z) \qquad 0\le x\le 1,\ 0\le y\le 1,\ 0\le z\le 1$$

$$f(x,y) = \frac{2}{3}\left[x+y+\frac{1}{2}\right] \qquad 0\le x\le 1,\ 0\le y\le 1$$

$$f(x) = \frac{2}{3}\left[x+1\right] \qquad 0\le x\le 1$$

(a) $\Rightarrow$ $E[X] = \frac{2}{3}\int_0^1 x(x+1)\,dx = \frac{2}{3}\left[\frac{1}{3}+\frac{1}{2}\right] = \frac{2}{3}\frac{5}{6} = \frac{5}{9} = E[Y] = E[Z]$

$$E[X^2] = \frac{2}{3}\int_0^1 x^2(x+1)\,dx = \frac{2}{3}\left[\frac{1}{4}+\frac{1}{3}\right] = \frac{2}{3}\frac{7}{12} = \frac{7}{18}$$

$$VAR[X] = \frac{7}{18} - \left(\frac{5}{9}\right)^2 = \frac{7}{18} - \frac{25}{81} = \frac{63-50}{162} = \frac{13}{162} = VAR[X] = VAR[Z]$$

$$E[XY] = \frac{2}{3}\int_0^1\int_0^1 xy\left(x+y+\tfrac{1}{2}\right)dx\,dy = \frac{2}{3}\left[\int_0^1\int_0^1 \left(x^2y + xy^2 + \tfrac{1}{2}xy\right)dx\,dy\right]$$

$$= \frac{2}{3}\int_0^1 dx\left[x^2\frac{1}{2} + x\frac{1}{3} + \frac{1}{2}x\frac{1}{2}\right] = \frac{2}{3}\left[\frac{1}{2}\frac{1}{3} + \frac{1}{3}\frac{1}{2} + \frac{1}{4}\frac{1}{2}\right]$$

$$\frac{4+4+3}{24}$$

$$= \frac{11}{36}$$

$$COV(X,Y) = \frac{11}{36} - \left(\frac{5}{9}\right)^2 = \frac{99-100}{324} = \frac{-1}{324}$$ almost uncorrelated

$$= COV(X,Z) = COV(Y,Z)$$

The optimum linear estimator for Y given X and Z is:

$$\hat{X} = (a_1, a_2)\begin{bmatrix} X-m_X \\ Z-m_Z \end{bmatrix} + m_Y$$

where from Eqn 6.63a

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \underbrace{\begin{bmatrix} VAR(X) & COV(X,Z) \\ COV(X,Z) & VAR(Z) \end{bmatrix}^{-1}}_{K_{XZ}^{-1}} \underbrace{\begin{bmatrix} COV(Y,X) \\ COV(Y,Z) \end{bmatrix}}_{E\left[(Y-m_Y)\begin{bmatrix} X-m_X \\ Z-m_Z \end{bmatrix}\right]}$$

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(6.73)

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \left[\frac{1}{324}\begin{bmatrix} 26 & -1 \\ -1 & 26 \end{bmatrix}\right]\begin{bmatrix} -1/324 \\ -1/324 \end{bmatrix} = \frac{324}{705}\begin{bmatrix} 26 & 1 \\ 1 & 26 \end{bmatrix}\begin{bmatrix} -\frac{1}{324} \\ -\frac{1}{324} \end{bmatrix}$$

$$= -\begin{bmatrix} \frac{27}{705} \\ \frac{27}{705} \end{bmatrix}$$

$$\hat{X}_{LMMSE} = -\frac{27}{705}[1,1]\begin{bmatrix} X-\frac{5}{9} \\ Z-\frac{5}{9} \end{bmatrix} + \frac{5}{9}$$

$$= -\frac{27}{705}\left(X-\frac{5}{9}\right) - \frac{27}{705}\left(Z-\frac{5}{9}\right) + \frac{5}{9}$$

$$= -\frac{27}{705}(X+Z) + \frac{5}{9}\left(\frac{651}{705}\right)$$

fm Eq 6.63)

$$MSE_{LMSE} = \sigma_X^2 - a^T\begin{bmatrix} cov(YX) \\ cov(YZ) \end{bmatrix} = \frac{13}{162} - \frac{27}{705}(1,1)\begin{bmatrix} -\frac{1}{324} \\ -\frac{1}{324} \end{bmatrix}$$

$$= \frac{13}{162} - \frac{54}{324(705)} = \frac{13}{162} - \frac{1}{235(6)} = 0.0795$$

(b)

$$f(y|x,z) = \frac{\frac{2}{3}(x+y+z)}{\frac{2}{3}(x+z+\frac{1}{2})} \qquad 0<y<1$$

$$E[Y|x,z] = \frac{1}{x+z+\frac{1}{2}}\int_0^1 y(x+y+z)\,dy = \frac{x\frac{1}{2}+\frac{1}{3}+z\frac{1}{2}}{x+z+\frac{1}{2}} = \frac{\frac{1}{2}(x+z)+\frac{1}{3}}{x+z+\frac{1}{2}}$$

$$\hat{Y}_{MMSE} = \frac{\frac{1}{2}(X+Z)+\frac{1}{3}}{X+Z+\frac{1}{2}}$$

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(b) continued —

6.73 Same equations following Eqn 6.59

$$MSE_{MMSE} = \int_0^1 \int_0^1 dx\,dz\; E[(Y-\hat{Y})^2 | x,z]\, f_{XZ}(x,z)$$

$$E[(Y-\hat{Y})^2 | x,z] = \int_0^1 (y - E[Y|x,z])^2 \frac{(x+y+z)}{x+z+\frac{1}{2}}\, dy$$

$$= E[Y^2|x,z] - 2E[Y|x,z] + E[Y|x,z]^2$$

$$E[Y^2|x,z] = \int_0^1 y^2 \frac{x+y+z}{x+z+\frac{1}{2}}\, dy = \frac{\frac{1}{3}x + \frac{1}{4} + \frac{1}{3}z}{x+z+\frac{1}{2}}$$

$$E[(Y-\hat{Y})^2|x,z] = \frac{\frac{1}{3}(x+z)+\frac{1}{4}}{x+z+\frac{1}{2}} - \left(\frac{\frac{1}{2}(x+z)+\frac{1}{3}}{x+z+\frac{1}{2}}\right)^2$$

$$= \frac{\frac{1}{12}\left[(x+z)^2 + (x+z) + \frac{1}{6}\right]}{\left((x+z)+\frac{1}{2}\right)^2}$$

$$E[(Y-\hat{Y})^2] = \frac{1}{12}\int_0^1\int_0^1 \frac{(x+z)^2+(x+z)+\frac{1}{6}}{\left((x+z)+\frac{1}{2}\right)^2} \frac{2}{3}\left((x+z)+\frac{1}{2}\right) dx\,dz$$

$$= \frac{1}{18}\int_0^1\int_0^1 \frac{(x+z)^2+(x+z)+\frac{1}{6}}{(x+z)+\frac{1}{2}}\, dx\,dz$$

$$= \frac{1}{18}\int_0^1\int_0^1 \left((x+z)+\frac{1}{2} - \frac{1/12}{(x+z)+\frac{1}{2}}\right) dx\,dz$$

$$= \frac{1}{18}\int_0^1 dx \left[x + \frac{1}{2} + \frac{1}{2} - \frac{1}{12}\ln(z+x+\tfrac{1}{2})\Big|_0^1 \right]$$

$\ln(x+\frac{3}{2}) - \ln(x+\frac{1}{2})$

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(b) continued—

6.73

$$MSE_{MMSE} = \frac{1}{18}\left[\frac{1}{2} + 1 - \frac{1}{12}\left\{\left[\left(x+\tfrac{3}{2}\right)\ln\left(x+\tfrac{3}{2}\right) - x\right]_0^1 - \left[\left(x+\tfrac{1}{2}\right)\ln\left(x+\tfrac{1}{2}\right) - x\right]_0^1\right\}\right]$$

$$\left(\tfrac{5}{2}\ln\tfrac{5}{2} - 1 - \tfrac{3}{2}\ln\tfrac{3}{2}\right) - \left(\tfrac{3}{2}\ln\tfrac{3}{2} - 1 - \tfrac{1}{2}\ln\tfrac{1}{2}\right)$$

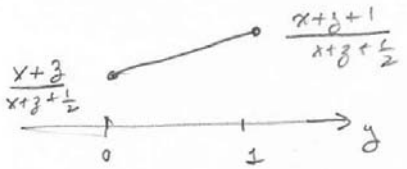
$$\tfrac{5}{2}\ln\tfrac{5}{2} - 2\tfrac{3}{2}\ln\tfrac{3}{2} + \tfrac{1}{2}\ln\tfrac{1}{2}$$

$$= \frac{1}{18}\left[\frac{3}{2} - \frac{1}{12}\left(\tfrac{5}{2}\ln\tfrac{5}{2} - 3\ln\tfrac{3}{2} + \tfrac{1}{2}\ln\tfrac{1}{2}\right)\right]$$

$= 0.08187$ larger than LMS MSE (need to recheck both)

(c) MAP Estimator

$$f(y|x,z) = \frac{(x+y+z)}{(x+z+\frac{1}{2})} \qquad 0 < y < 1$$

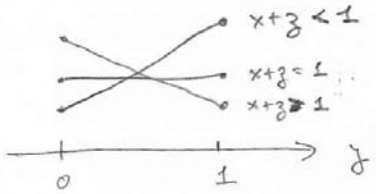


$$\Rightarrow \hat{Y}_{MAP} = 1 \qquad MSE_{MAP} = E[(Y-1)^2] = E[Y^2] - 2E[Y] + 1$$

$$= \frac{7}{18} - 2\left(\frac{5}{9}\right) + 1 = .277$$

ML Estimator

$$f(x,z|y) = \frac{\frac{2}{3}(x+y+z)}{\frac{2}{3}(y+1)}$$



$$\therefore \hat{Y}_{ML} = \begin{cases} 1 & x+z<1 \\ 0 & x+z>1 \end{cases}$$

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6.73

(d) $$MSE_{ML} = \iiint_{x+z<1} (y-1)^2 f(x,y,z)\,dx\,dy\,dz + \iiint_{x+z>1} y^2 f(x,y,z)\,dx\,dy\,dz$$

$$= \int_0^1 dx \int_0^{1-x} dz \int_0^1 (y-1)^2 \tfrac{2}{3}(x+y+z)\,dy + \int_0^1 dx \int_{1-x}^1 dz \int_0^1 y^2 \tfrac{2}{3}(x+y+z)\,dy$$

$$= \tfrac{2}{3}\int_0^1 (y-1)^2 dy \int_0^1 dx \underbrace{\int_0^{1-x} dz\,(x+y+z)}_{x(1-x)+y(1-x)+(1-x)^2}$$

$$+ \tfrac{2}{3}\int_0^1 y^2 dy \int_0^1 dx \underbrace{\int_{1-x}^1 dz\,(x+y+z)}_{x(x)+yx+\frac{1}{2}(1-(1-x)^2)}$$

$$= \tfrac{2}{3}\int_0^1 dy\,(y-1)^2\left(1+y-y\tfrac{1}{2}-\tfrac{1}{2}\right) + \tfrac{2}{3}\int_0^1 dy\,y^2\left(\tfrac{2}{3}+\tfrac{1}{2}y\right)$$

$$= \frac{5}{36} + \frac{17}{54} = \frac{15+34}{108} = \frac{49}{108} = .453$$

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6.74

$$E[X_1] = \int_0^1 x_1\,dx_1 = \frac{1}{2} \qquad E[X_1^2] = \int_0^1 x_1^2\,dx_1 = \frac{1}{3} \qquad \text{VAR}[X_1] = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$E[X_2] = E[\underbrace{E[X_2|X_1]}_{\frac{1}{2}X_1}] = \frac{1}{2}E[X_1] = \frac{1}{4}$$ given X_1, X_2 uniform in $[0, X_1]$

$$E[X_3] = E[\underbrace{E[X_3|X_2]}_{\frac{1}{2}X_2}] = \frac{1}{2}E[X_2] = \frac{1}{8}$$

$$E[X_2^2] = E[\underbrace{E[X_2^2|X_1]}_{\frac{X_1^2}{3}}] = \frac{1}{3}E[X_1^2] = \frac{1}{9} \qquad \text{VAR}[X_2] = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}$$

$$E[X_3^2] = E[\tfrac{1}{3}X_2^2] = \frac{1}{3}E[X_2^2] = \frac{1}{27} \qquad \text{VAR}[X_3] = \frac{1}{27} - \frac{1}{64} = \underbrace{\frac{37}{27(64)}}_{1728}$$

$$E[X_1X_2] = E[\underbrace{XE[X_2|X_1]}_{\frac{1}{2}X_1}] = \frac{1}{2}E[X_1^2] = \frac{1}{6}$$

$$E[X_1X_3] = E[X_1\underbrace{E[\underbrace{E[X_3|X_2X_1]}_{\frac{1}{2}X_2}|X_1]}_{\frac{1}{4}X_1}] = \frac{1}{4}E[X_1^2] = \frac{1}{12}$$

$$E[X_2X_3] = E[X_2\underbrace{E[X_3|X_2]}_{\frac{1}{2}X_2}] = \frac{1}{2}E[X_2^2] = \frac{1}{18}$$

(a) Best Linear MSE Estimate

$$\underset{LMSE}{\hat{X}_2} = (a_1, a_2)\begin{bmatrix} X_1 - m_1 \\ X_3 - m_3 \end{bmatrix} + m_2 = -1.696\left(X_1 - \tfrac{1}{2}\right) + 2.786\left(X_2 - \tfrac{1}{8}\right) + \tfrac{1}{4}$$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{12} & \frac{1}{12} - \frac{1}{2}\frac{1}{8} \\ \frac{1}{48} & \frac{37}{1728} \end{bmatrix}^{-1}\begin{bmatrix} \frac{1}{6} - \frac{1}{2}\frac{1}{4} \\ \frac{1}{18} - \frac{1}{4}\frac{1}{8} \end{bmatrix} = \begin{bmatrix} -1.696 \\ 2.786 \end{bmatrix}$$

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6.74

$$f(x_1,x_2,x_3) = \frac{1}{x_1 x_2} \qquad 0 < x_3 < x_2 < x_1 < 1$$

$$f(x_1,x_3) = \int_{x_3}^{x_1} \frac{1}{x_1 x_2}\, dx_2 = \frac{1}{x_1}\left[\ln x_1 - \ln x_3\right] = \frac{1}{x_1}\ln\frac{x_1}{x_3}$$

$$f(x_2|x_1,x_3) = \frac{\frac{1}{x_1 x_2}}{\frac{1}{x_1}\ln\frac{x_1}{x_3}} = \frac{1/x_2}{\ln x_1/x_3} \qquad x_3 < x_2 < x_1$$

decreases with increasing x_2

$$E[X_2|x_1,x_3] = \frac{1}{\ln\frac{x_1}{x_3}}\int_{x_3}^{x_1} x_2 \frac{1}{x_2}\, dx_2 = \frac{x_1 - x_3}{\ln(x_1/x_3)}$$

(b)

$$\hat{X}_{2\,MMSE} = \frac{x_1 - x_3}{\ln(x_1/x_3)}$$

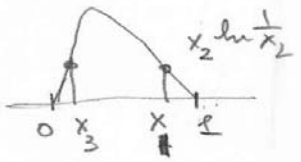
$$\hat{X}_{2\,MAP} = \frac{1/x_3}{\ln x_1/x_3}$$

from Prob. 6.13 : $f(x_2) = -\ln x_2$

(c)

$$f(x_1,x_3|x_2) = \frac{\frac{1}{x_1 x_2}}{-\ln x_2} = \frac{1/x_1}{x_2 \ln\frac{1}{x_2}}$$

$0 < x_3 < x_2 < x_1 < 1$



$f(x_1,x_3|x_2)$ is maximized at either x_3 or x_1 according to whether $x_3 \ln\frac{1}{x_3} \lessgtr x_1 \ln\frac{1}{x_1}$

$$\hat{X}_{2\,ML} = \begin{cases} x_3 & \text{if } x_3 \ln\frac{1}{x_3} < x_1 \ln\frac{1}{x_1} \\ x_1 & \text{if } x_3 \ln\frac{1}{x_3} > x_1 \ln\frac{1}{x_1} \end{cases}$$

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(6.74) (d)

$$\hat{X}_3 = (a_1, a_2)\begin{bmatrix} X_1 - m_1 \\ X_2 - m_2 \end{bmatrix} + m_3$$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{12} & \frac{1}{6} - \frac{1}{2}\frac{1}{4} \\ \frac{1}{24} & \frac{7}{144} \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{12} - \frac{1}{2}\frac{1}{8} \\ \frac{1}{18} - \frac{1}{4}\frac{1}{8} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix}$$

$$\therefore \quad \hat{X}_3 = \frac{1}{2}\left(X_2 - \frac{1}{4}\right) + \frac{1}{8} = \frac{1}{2}X_2$$

$\hat{X}_3$ LMSE

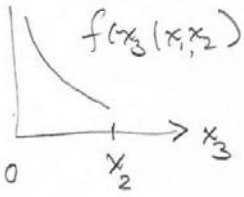
Correlation structure captures fact that X_3 does not depend on X_1 when X_2 is known!

$$f(x_3 | x_1, x_2) = \frac{\frac{1}{x_1 x_2}}{\frac{1}{x_1}} = \frac{1}{x_2} \qquad 0 < x_3 < x_2$$

$$E[X_3 | X_1, X_2] = \int_0^{x_2} x_3 \cdot \frac{1}{x_2}\, dx_3 = \frac{1}{x_2}\frac{x_2^2}{2} = \frac{x_2}{2}$$

$\hat{X}_{3_{MMSE}} = \frac{1}{2}X_2$ same as linear estimator

$\hat{X}_{3_{MAP}} = 0$



$$f(x_1, x_2 | x_3) = \frac{\frac{1}{x_1 x_2}}{\frac{1}{2}(\ln x_3)^2} \qquad \text{for } x_3 < x_2 < x_1 < 1$$

the pdf is maximized wrt x_3 when the denominator is minimized. This occurs at $x_3 = x_2$

$$\hat{X}_{3_{ML}} = X_2$$

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6.75

(a) $$f(\underline{y}|\underline{b}) = \frac{1}{\sqrt{2\pi}^k} e^{-\frac{1}{2}\sum_{j=1}^{k}(y_i - \alpha_i b_i)^2}$$

$$P[\underline{b}] = \left(\frac{1}{2}\right)^k$$

$$f(\underline{y}) = \sum_{\underline{b}} f(\underline{y}|\underline{b}) P[\underline{b}] = \frac{1}{2^k} \sum_{\underline{b}} f(\underline{y}|\underline{b})$$

ML Estimator maximizes

$$\max_{\underline{b}} f(\underline{y}|\underline{b}) = \max_{\underline{b}} \frac{1}{\sqrt{2\pi}^k} e^{-\frac{1}{2}\sum_{j=1}^{k}(y_i - \alpha_i b_i)^2}$$

which is equivalent to minimizing the exponent

$$\min_{\underline{b}} \sum_{j=1}^{k} (y_i - \alpha_i b_i)^2 \Rightarrow \text{find } A\underline{b} \text{ that is closest to } \underline{Y} \text{ in Euclidean distance.}$$

MAP Estimator is same as ML estimator since $\underline{b}$ are equally probable.

(b) This problem involves finding a estimator for a vector of random variables $\underline{b}$ based on a vector of observations $\underline{Y}$. For linear MSE estimators we generalize Eqs 6.63ab; for $\underline{X} = \underline{b}$:

$$\hat{\underline{X}} = A^T(\underline{Y} - \underline{m}_Y) + \underline{m}_{\underline{X}}$$

where

$$A = K_Y^{-1} K_{XY} \quad \text{where } K_{XY} = E[(\underline{X} - \underline{m}_X)(\underline{Y} - \underline{m}_Y)^T]$$

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6.75

$$K_{\underline{bY}} = E[(\underline{b} - \underline{m}_b)(\underline{Y} - \underline{m}_Y)^T]$$

$\underline{m}_b = \underline{0}$ since b_i equally likely to be ± 1.

$$E[\underline{b}(\underline{Y} - \underline{m}_Y)^T] = \left[E[B_i(Y_j - m_j)] \right]$$

Recall $Y_j = \alpha_j B_j + N_j$

$$\therefore \quad E[B_i(Y_j - m_j)] = \begin{cases} E[B_i(Y_i - m_i)] & i = j \\ E[B_i]E[(Y_j - m_j)] = 0 & i \neq j \end{cases}$$

$$E[B_i(Y_i - m_i)] = E[B_i(Y_i - m_i) \mid B_i = +1]\tfrac{1}{2} + E[B_i(Y_i - m_i) \mid B_i = -1]\tfrac{1}{2}$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} (y_i - m_i) e^{-(y-\alpha_i)^2/2} dy - \frac{1}{2}\int_{-\infty}^{\infty} (y_i - m_i) e^{-(y+\alpha_i)^2/2} dy$$

$$= \tfrac{1}{2}[\alpha_i - m_i] - \tfrac{1}{2}[-\alpha_i - m_i]$$

$$= \alpha_i$$

$$\therefore \quad K_{\underline{bY}} = \begin{bmatrix} \alpha_1 & & 0 \\ & \ddots & \\ 0 & & \alpha_K \end{bmatrix} = A$$

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$$K_Y = E\left[(\underline{Y}-\underline{m}_Y)(\underline{Y}-\underline{m}_Y)^T\right]$$

$$= E\left[\; E\left[(Y_i-m_i)(Y_j-m_j)\right]\right]$$

$$E\left[(Y_i-m_i)(Y_j-m_j)\right] = \begin{cases} E\left[(Y_i-m_i)^2\right] & i=j \\ E\left[Y_i-m_i\right]E\left[(Y_j-m_j)\right]=0 & i\neq j \end{cases}$$

$$m_i = E[Y_i] = \underbrace{E[Y_i \mid B_i=+1]}_{+\alpha_i}\frac{1}{2} + \underbrace{E[Y_i \mid B_i=-1]}_{-\alpha_i}\frac{1}{2} = 0$$

$$E\left[(Y_i-m_i)^2\right] = E[Y_i^2]$$

$$= E[Y_i^2 \mid B_i=+1]\frac{1}{2} + E[Y_i^2 \mid B_i=-1]\frac{1}{2}$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} y^2 e^{-(y-\alpha_i)^2/2}\,dy + \frac{1}{2}\int_{-\infty}^{\infty} y^2 e^{-(y+\alpha_i)^2/2}$$

$$= \frac{1}{2}\left(1+\alpha_i^2\right) + \frac{1}{2}\left(1+(-\alpha_i)^2\right)$$

$$= 1+\alpha_i^2$$

$$K_Y = \begin{bmatrix} 1+\alpha_1^2 & & 0 \\ & 1+\alpha_2^2 & \\ 0 & & \ddots \; 1+\alpha_k^2 \end{bmatrix}$$

$$\hat{b}_{LMSE} = \begin{bmatrix} \frac{1}{1+\alpha_1^2} & & 0 \\ & \frac{1}{1+\alpha_2^2} & \\ 0 & & \ddots \; \frac{1}{1+\alpha_k^2} \end{bmatrix}\begin{bmatrix} \alpha_1 & & 0 \\ & \ddots & \\ 0 & & \alpha_k \end{bmatrix}\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_k \end{bmatrix}$$

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$$\hat{b}_{MLSE} = \begin{bmatrix} \frac{\alpha_1}{1+\alpha_1} Y_1 \\ \vdots \\ \frac{\alpha_K}{1+\alpha_K} Y_K \end{bmatrix}$$

The decision/estimate for each component is based solely on the corresponding observation. There is no "coupling" between different channel components.

6.76 From problem 6.58

$$E[\underline{Y}|\underline{b}] = A\underline{R}\underline{b} \Rightarrow E[\underline{Y}] = E[A\underline{R}\underline{b}] = A\underline{R}E[\underline{b}] = \underline{0}$$

Also

$$f(\underline{y}|\underline{b}) = \frac{\exp\left\{-\frac{1}{2}(\underline{y}-A\underline{R}\underline{b})^T(\underline{y}-A\underline{R}\underline{b})\right\}}{(2\pi)^{K/2}} =$$

$$= \frac{\exp\left\{-\frac{1}{2}(\underline{y}-\underline{b}')^T(\underline{y}-\underline{b}')\right\}}{(2\pi)^{K/2}} \quad \text{where } \underline{b}' = A\underline{R}\underline{b}$$

$$= \frac{\exp\left\{-\frac{1}{2}\sum_{j=1}^{K}(y_j-b_j')^2\right\}}{(2\pi)^{K/2}}$$

This has the same form as the pdf in problem 6.75

The ML estimator minimizes the exponent by finding

$$\min_{\underline{b}'} \sum_{j=1}^{K}(y_i-b_i')^2 \Rightarrow \text{find } \underline{b}' \text{ that is closest to observation } \underline{Y} \text{ in Euclidean distance.}$$

MAP estimator is same as ML estimator.

The estimate for $\underline{b}$ is obtained from

$$\hat{\underline{b}}_i = (AR)^{-1}\underline{b}_i'$$

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6.76

$$K_{\underline{b}\underline{Y}} = E[(\underline{b} - \underline{m}_{\underline{b}})(\underline{Y} - \underline{m}_{\underline{Y}})^T] = E[\underline{b}\underline{Y}^T]$$

($\underline{m}_{\underline{b}} = \underline{0}$)

$$= E[\underline{b}(A R \underline{b} + \underline{N})^T]$$

$$= E[\underline{b}(\underline{b}^T R^T A^T + \underline{N}^T)]$$

$$= \underbrace{E[\underline{b}\underline{b}^T]}_{I} R^T A^T + \underbrace{E[\underline{b}]E[\underline{N}^T]}_{0}$$

$$= R^T A^T$$

$$K_{\underline{Y}} = E[\underline{Y}\underline{Y}^T] = E[(A R \underline{b} + \underline{N})(A R \underline{b} + \underline{N})^T]$$

$$= A R E[\underline{b}\underline{b}^T] R^T A^T + \underbrace{E[\underline{N}\underline{N}^T]}_{I}$$

$$= A R R^T A^T + I$$

since R is symmetric

$$= A R^2 A + I$$

$$\hat{\underline{b}}_{MMSE} = (A R^2 A + I)^{-1} R^T A^T \underline{Y}$$

This expression generalizes the result for estimating a signal from a noisy observation in Prob. 6.72.

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6.77

From Eqns. 4.84, letting D correspond to X_1 and B to X_2, the best coefficients are:

a)
$$a = \frac{\sigma_B^2 COV(D,E) - COV(B,D)COV(B,E)}{\sigma_D^2\sigma_B^2 - COV(B,D)^2}$$
$$= \frac{\sigma^2(\sigma^2\rho) - \sigma^2\rho^2\sigma^2\rho}{\sigma^4 - \sigma^4\rho^4} = \frac{\rho - \rho^3}{1-\rho^4} = \frac{\rho}{1+\rho^2}$$

and

$$b = \frac{\sigma_D^2 COV(B,E) - COV(B,D)COV(D,E)}{\sigma_D^2\sigma_B^2 - COV(B,D)^2}$$
$$= \frac{\sigma^2(\sigma^2\rho) - \sigma^2\rho^2(\sigma^2\rho)}{\sigma^4 - \sigma^4\rho^4} = \frac{\rho}{1+\rho^2}$$

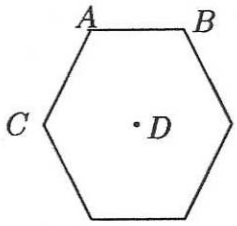
b)
$$\mathcal{E}[(E - \underbrace{(aD+bB)}_{\hat{E}})^2] = \mathcal{E}[(E-\hat{E})(E-aD-bB)]$$
$$= \mathcal{E}[(E-\hat{E})E] - a\underbrace{E[(E-\hat{E})D]}_{0} - b\underbrace{\mathcal{E}[(E-\hat{E})B]}_{0}$$
since error and observations are orthogonal
$$= \mathcal{E}[(E-aD-bB)E]$$
$$= \mathcal{E}[E^2] - a\mathcal{E}[DE] - b\mathcal{E}[BE]$$
$$= \sigma^2 - \frac{\rho}{1+\rho^2}\rho\sigma^2 - \frac{\rho}{1+\rho^2}\rho\sigma^2$$
$$= \sigma^2 - \frac{2\rho^2}{1+\rho^2}\sigma^2 = \sigma^2\left\{1 - \frac{2\rho^2}{1+\rho^2}\right\}$$

6.78

$$\mathcal{E}[(X_3 - aX_1 - bX_2)^2] = \mathcal{E}[X_3(X_3 - aX_1 - bX_2)]$$
$$-a\underbrace{\mathcal{E}[X_1(X_3 - aX_1 - bX_2)]}_{0} - b\underbrace{\mathcal{E}[X_2(X_30aX_1 - bX_2)]}_{0}$$
since error and observations are orthogonal
$$= \mathcal{E}[X_3^2] - a\mathcal{E}[X_1X_3] - b\mathcal{E}[XX_3X_2]$$
$$= \sigma^2 - \frac{\rho_2 - \rho_1^2}{1-\rho_1^2}\rho_2\sigma^2 - \frac{\rho_1(1-\rho_2)}{1-\rho_1^2}\rho_1\sigma^2$$
$$= \sigma^2 - \frac{\rho_2^2 - \rho_1^3\rho_2 + \rho_1^2 - \rho_1^3\rho_2}{1-\rho_1^2}\sigma^2$$
$$= \sigma^2 - \frac{\rho_2^2 - 2\rho_1^2\rho_2 + \rho_1^4 + \rho_1^2 - \rho_1^4}{1-\rho_1^2}\sigma^2$$
$$= \sigma^2\left\{1 - \rho_1^2 - \frac{(\rho_1^2 - \rho_2)^2}{1-\rho_1^2}\right\} \quad \surd$$

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6.79



The performance of using $\{A, C\}$ or $\{A, B\}$ is the same.

a) Using $\{A, B\}$ assuming $E[X_i] = 0$

$\cdot x_1$ $\cdot x_3$

$\cdot x_3$

$$\begin{aligned} a &= \frac{1 \cdot p - p \cdot p}{1 \cdot 1 - p^2} = \frac{p}{1+p} \\ b &= \frac{p}{1+p} \\ c &= E[(X_3 - aX_1 - bX_2)^2] \\ &= E[X_3^2] + a^2 E[X_1^2] + b^2 E[X_2^2] - 2aE[X_1 X_3] + 2abE[X_1 X_2] \\ &= 1 + a^2 + b^2 - 2ap - 2bp + 2abp \\ &= \frac{1 - p - 2p^2}{1+p} \end{aligned}$$

b) Using $\{B, C\}$, assuming $e[X_i] = 0$

$\cdot x_1$

$\cdot x_1$ $\cdot x_3$

$$\begin{aligned} a &= \frac{1 \cdot p - p^{\sqrt{e}} \cdot p}{1 \cdot 1 - p^{2\sqrt{3}}} = \frac{p}{1 + p^{\sqrt{3}}} \\ b &= \frac{p}{1 + p^{\sqrt{3}}} \\ e &= E[(X_3 - aX_1 - bX_2)^2] \\ &= E[X_3^2] + a^2 E[X_1^2] + b^2 E[X_2^2] - 2aE[X_1 X_3] - 2abe[X_2 X_3] + 2abe[X_2 X - 3] \\ &= 1 + a^2 + b^2 - 2ap - 2bp + 2abp^{\sqrt{3}} \\ &= \frac{1 - p^{\sqrt{3}} - 2p^2}{1 + p^{\sqrt{3}}} \end{aligned}$$

$$\frac{1 - p - 2p^2}{1+p} > \frac{1 - p^{\sqrt{3}} - 2p^2}{1 + p^{\sqrt{3}}} \quad \text{for } 0 < p < 1$$

We should use samples B and C to give a smaller predicttion error.

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6.6 Generating Correlated Vector Random Variables

6.80

4.96 $K = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$

$$\begin{aligned} \det|K - \lambda I| &= \lambda^2 - 6\lambda + 7 \\ \lambda_1, \lambda_2 &= 3 \pm \sqrt{2} \quad \text{eigenvalues} \end{aligned}$$

The orthonormal eigenvectors are:

$$\underline{e}_1 = \frac{1}{\sqrt{4+2\sqrt{2}}}\begin{bmatrix} 1 \\ 1+\sqrt{2} \end{bmatrix} \qquad \underline{e}_2 = \frac{1}{\sqrt{4-2\sqrt{2}}}\begin{bmatrix} 1 \\ 1-\sqrt{2} \end{bmatrix}$$

$$P = [\underline{e}_1, \underline{e}_2] = \begin{bmatrix} .38268 & .92388 \\ .92388 & -.38268 \end{bmatrix}$$

$$A = PD^{1/2} = \begin{bmatrix} .80401 & 1.16342 \\ 1.94107 & -.48190 \end{bmatrix}$$

$$A = \begin{bmatrix} .80401 & 1.16342 \\ 1.94107 & -.48190 \end{bmatrix}$$

Check

$$AA^+ = \begin{bmatrix} .80401 & 1.16342 \\ 1.94107 & -.48190 \end{bmatrix}\begin{bmatrix} .80401 & 1.94107 \\ 1.16342 & -.48190 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \quad \surd$$

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6.81

$$K = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$$

Using `[P, D] = eig(K)` we obtain

$$P = \begin{bmatrix} -0.92388 & 0.38268 \\ 0.38268 & 0.92388 \end{bmatrix} \qquad D = \begin{bmatrix} 1.58579 & 0 \\ 0 & 4.41421 \end{bmatrix}$$

then

$$A = (P * \text{sqrt}(D))' = \begin{bmatrix} -1.16342 & 0.48191 \\ 0.80402 & 1.94107 \end{bmatrix}$$

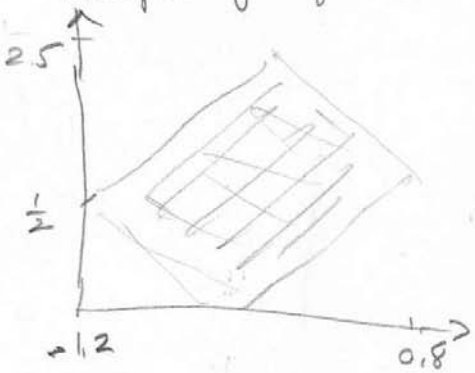
The desired transform is then

$$\underline{Y} = A'\underline{X}$$

(a) To generate pairs with covariance K where $\underline{X}$ are iid uniform pairs for $[0, 1]$:

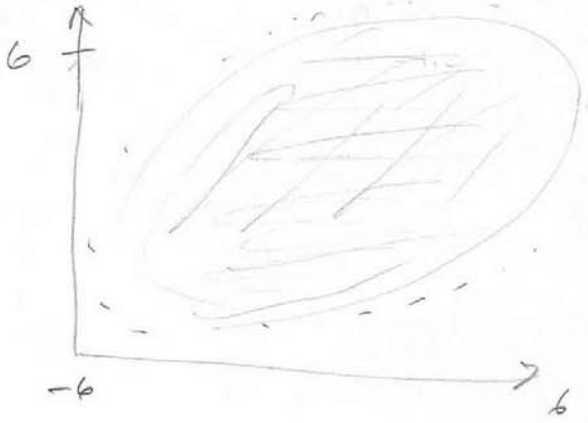
```
x = uniform_rnd(0, 1, 2, 1000);
z = A' * x
plot( z(1, :), z(2, :), '+' )
```

1000 pairs of uniform iid RVs



(b)

```
x = normal_rnd(0, 1, 2, 100)
z = A' * x
plot (z(1, :), z(2, :), '+')
```



jointly Gaussian RVs

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6.82

$$\underline{m}_X = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \qquad K_Y = \begin{bmatrix} 3/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 3/2 \end{bmatrix}$$

$[P, D] = eig(K)$ gives $P = \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -1 & 0 & 0 \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$

First we need to generate X: $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$A = (P * sqrt(D))'$

$$A' = \begin{bmatrix} 0 & 1/\sqrt{2} & 1 \\ -1 & 0 & 0 \\ 0 & -1/\sqrt{2} & 1 \end{bmatrix}$$

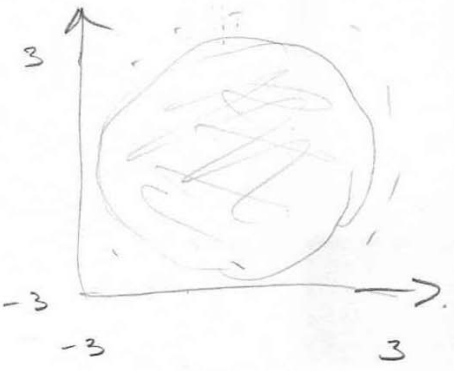
```
X = normal_rnd(0,1,3,1000)
z = A' * X
plot(z(1,:), z(3,:), '+')
```



plots X_1, X_3 scattergram

Next we need to obtain iid zero-mean unit variance Gauss pair:

```
w = Az
plot(w(1,:), w(2,:), '+')
```



This is somewhat artificial since $AA^T = I$; but we cannot avoid the need to generate $\underline{X}$.

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6.83 Let A be such that

$$A^T K_x A = \Lambda \Rightarrow K_x = A \Lambda A^T$$

Consider

$$\underline{Y} = A^{-1}(\underline{x} - \underline{m}_x)$$

$$E[\underline{Y}] = A^{-1} E[\underline{x} - \underline{m}_x] = \underline{0}$$

$$K_Y = E[\underline{Y}\,\underline{Y}^T]$$

$$= E[A^{-1}(\underline{x} - \underline{m}_x)(\underline{x} - \underline{m}_x)^T A^{-1T}]$$

$$= A^{-1} K_x A^{-1T}$$

$$= A^{-1} A \Lambda A^T (A^T)^{-1}$$

$$= \Lambda = \text{diag}[\lambda_i]$$

$$\therefore f_Y(\underline{y}) = \frac{1}{(\sqrt{2\pi})^n} \exp\left\{-\frac{1}{2}\sum_{i=1}^{n} y_i^2/2\lambda_i\right\}$$

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6.84

(a) $\mathcal{E}[Y_k] = \frac{1}{2}\mathcal{E}[X_k] + \frac{1}{2}\mathcal{E}[X_{k-1}] = 0$

$$\begin{aligned} COV(Y_k Y_{k'}) &= \mathcal{E}[Y_k Y_{k'}] = \frac{1}{4}\mathcal{E}[(X_k + X_{k-1})(X_{k'} + X_{k'-1})] \\ &= \frac{1}{4}\mathcal{E}[X_k X_{k'} + X_k X_{k'-1} + X_{k-1} X_{k'} + X_{k-1} X_{k'-1}] \end{aligned}$$

Since the X_k's are independent, the above terms are all zero except when $k = k'$ or $k = k'-1$ or $k = k'+1$. Then

$$COV(Y_k Y_{k'}) = \begin{cases} \frac{1}{4}\mathcal{E}[X_k^2 + X_{k-1}^2] = \frac{1}{2} & k = k' \\ \frac{1}{4}\mathcal{E}[X_{k-1}^2] = \frac{1}{4} & k' = k-1 \\ \frac{1}{4}\mathcal{E}[X_k^2] = \frac{1}{4} & k' = k+1 \end{cases}$$

$$\therefore \quad K = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & 0 & 0 & \cdots & & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & \cdots & & \vdots \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & \cdots & \\ & & & \ddots & & & 0 \\ & & & & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ & 0 & 0 & & \cdots & 0 & \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

where
$COV(X_1, X_1) = \frac{1}{4}$
$COV(X_n, X_n) = \frac{1}{4}$

(b) The following Octave code generates a sequence of 1000 samples:

```
X = normal_rnd(0, 1, 1, 1000);
Y = (X + [0 X(1:length(X)-1)])./2;
cov(Y);
```

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6.85

$$\begin{aligned}
\mathcal{E}[Y-K] &= \mathcal{E}[X_k - X_{k-1}] = 0 \\
COV(Y_k, Y_{k'}) &= \mathcal{E}[(X_k - X_{k-1})(X_{k'} - X_{k'-1})] \\
&= \mathcal{E}[X_k X_{k'}] - \mathcal{E}[X_k X_{k'-1}] - \mathcal{E}[X_{k-1} X_{k'}] + \mathcal{E}[X_{k-1} X_{k'-1}] \\
COV(Y_k Y_{k'}) &= 0 \text{ except when } k' = k, k-1, k+1 \text{ then} \\
COV(Y_k, Y_{k'}) &= \begin{cases} \mathcal{E}[X_k^2] + \mathcal{E}[X_{k-1}^2] = 2 & k = k' \\ -\mathcal{E}[X_k^2] = -1 & k' = k+1 \\ -\mathcal{E}[X_{k-1}^2] = -1 & k' = k-1 \end{cases} \\
COV(X_1, X_1) &= 1 = COV(X_n, X_n)
\end{aligned}$$

$$K = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & \cdots & & \\ -1 & 2 & -1 & 0 & 0 & \cdots & & \vdots \\ 0 & -1 & 2 & -1 & 0 & \cdots & & \\ & & \ddots & & \ddots & & & 0 \\ & & & & & 0 & -1 & 2 & -1 \\ 0 & 0 & & \cdots & & 0 & 0 & -1 & 1 \end{bmatrix}$$

6.86

Given $AK_X A^T = \Lambda$

Let $B = UA$ where U is an orthogonal matrix $UU^T = I$

$$\begin{aligned}
BK_X B^T &= UAK_X (UA)^T \\
&= UAK_X A^T U^T \\
&= U \Lambda U^T \\
&= \Lambda U U^T \\
&= \Lambda
\end{aligned}$$

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6.87

$$X = U_1$$
$$Y = U_1 + U_2$$
$$Z = U_1 + U_2 + U_3$$

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}_{A} \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix}$$

$$E[\underline{U}] = \underline{0} \qquad K_{\underline{U}} = I$$

$$K_{\underline{X}} = A K_{\underline{U}} A^T = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\text{Let } \Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} 0.30798 & 0 & 0 \\ 0 & 0.64310 & 0 \\ 0 & 0 & 5.04892 \end{bmatrix}$$

$$K_{\underline{X}} = A \Lambda^{-\frac{1}{2}} \Lambda \Lambda^{-\frac{1}{2}} A^T$$

$$= (A \Lambda^{-\frac{1}{2}}) \Lambda (\Lambda^{-\frac{1}{2}} A^T)$$

$$\Lambda^{1/2} A^{-1} K_X A^{-1T} \Lambda^{\frac{1}{2}} = \Lambda$$

$$\Lambda^{1/2} A^{-1} = \begin{bmatrix} \sqrt{\lambda_1} & 0 & 0 \\ 0 & \sqrt{\lambda_2} & 0 \\ 0 & 0 & \sqrt{\lambda_3} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{\lambda_1} & 0 & 0 \\ -\sqrt{\lambda_2} & \sqrt{\lambda_2} & 0 \\ 0 & -\sqrt{\lambda_3} & \sqrt{\lambda_3} \end{bmatrix}}_{\text{causal}}$$

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6.88

$$K_X = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{bmatrix} \qquad D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \qquad P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

(a)

$$\begin{bmatrix} a & 0 \\ b & c \end{bmatrix} \begin{bmatrix} 3/2 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix} \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} a & 0 \\ b & c \end{bmatrix} \begin{bmatrix} \frac{3}{2}a & \frac{3}{2}b - \frac{1}{2}c \\ -\frac{1}{2}a & -\frac{1}{2}b + \frac{3}{2}c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\frac{3}{2}a^2 = 1 \qquad a = \sqrt{\frac{2}{3}}$$

$$\frac{3}{2}ab - \frac{1}{2}ac = 0 \implies 3b = c.$$

$$\frac{3}{2}b^2 - \frac{1}{2}bc - \frac{1}{2}bc + \frac{3}{2}c^2 = 2$$

$$\frac{3}{2}b^2 - b(3b) + \frac{3}{2}(9b^2) = 2$$

$$12b^2 = 2 \implies b = \sqrt{\frac{1}{6}}$$

$$\implies c = 3\sqrt{\frac{1}{6}} = \sqrt{\frac{9}{6}} = \sqrt{\frac{3}{2}}$$

$$A = \begin{bmatrix} \sqrt{\frac{2}{3}} & 0 \\ \sqrt{\frac{1}{6}} & \sqrt{\frac{3}{2}} \end{bmatrix}$$

then $AK_XA^T = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ ✓

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6.88 (b)

$$K_X = \begin{bmatrix} 3/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 3/2 \end{bmatrix}$$

Consider the reduced matrix $\begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{bmatrix}$

Proceeding as in 6.88a

$$\begin{bmatrix} a & 0 \\ b & c \end{bmatrix}\begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{bmatrix}\begin{bmatrix} a & b \\ 0 & c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} a & 0 \\ b & c \end{bmatrix}\begin{bmatrix} \frac{3}{2}a & \frac{3}{2}b+\frac{1}{2}c \\ \frac{1}{2}a & \frac{1}{2}b+\frac{3}{2}c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\frac{3}{2}a^2 = 1 \qquad a = \sqrt{\frac{2}{3}}$$

$$\frac{3}{2}ab + \frac{1}{2}ac = 0 \Rightarrow -3b = c$$

$$\frac{3}{2}b^2 = b(3b) + \frac{3}{2}(9b^2) = 2$$

$$12b^2 = 2 \Rightarrow b = \sqrt{\frac{2}{12}} = \sqrt{\frac{1}{6}}$$

$$\Rightarrow c = -3\sqrt{\frac{1}{6}} = -\sqrt{\frac{9}{6}} = -\sqrt{\frac{3}{2}}$$

$$A = \begin{bmatrix} \sqrt{\frac{2}{3}} & 0 & 0 \\ 0 & 1 & 0 \\ \sqrt{\frac{1}{6}} & 0 & -\sqrt{\frac{3}{2}} \end{bmatrix}$$

then $AK_XA' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

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Problems Requiring Cumulative Knowledge

6.89

(a)

$$X_{n-1} = \tfrac{1}{2}U_{n-2} + \tfrac{1}{2}U_{n-1}$$

$$X_n = \tfrac{1}{2}U_{n-1} + \tfrac{1}{2}U_n$$

$$X_{n+1} = \tfrac{1}{2}U_n + \tfrac{1}{2}U_{n+1}$$

$$Z = U_{n+1}$$

U_n are iid zero-mean unit-variance Gauss RV

$$\underline{X}_a = \begin{bmatrix} X_{n-1} \\ X_n \\ X_{n+1} \\ Z \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} U_{n-2} \\ U_{n-1} \\ U_n \\ U_{n+1} \end{bmatrix}$$

$$E[\underline{X}_a] = AE[\underline{U}] = \underline{0}$$

$$K_{\underline{X}_a} = AK_{\underline{U}}A^T = AA^T = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 1 \end{bmatrix}$$

We are only interested in

$$\underline{X} = \begin{bmatrix} X_{n-1} \\ X_n \\ X_{n+1} \end{bmatrix}$$ which are jointly Gaussian with $E[\underline{X}] = \underline{0}$

and $$K_{\underline{X}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

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6.89 — continued —

$$X_{n+m} = \frac{1}{2}U_{n+m-1} + \frac{1}{2}U_{n+m}$$

$$X_{n+2m} = \frac{1}{2}U_{n+2m-1} + \frac{1}{2}U_{n+2m}$$

$$E[X_{n+2m}] = 0$$

$$VAR[X_{n+2m}] = \frac{1}{2}VAR(U_{n-1} + U_n) = \frac{1}{2}\underbrace{2VAR(U)}_{1} = 1$$

$\therefore$ X_n, X_{n+m}, X_{n+2m} are iid zero-mean, unit variance Gaussian RVs.

(b)

$$\underline{X}_a = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} U_{n-2} \\ U_{n-1} \\ U_n \\ U_{n+1} \end{bmatrix}$$

$$E[\underline{X}_a] = \underline{0}$$

$$K_{\underline{X}_a} = AK_UA^T = AA^T = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} & 0 & 0 \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} & 0 \\ 0 & -\frac{1}{4} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 1 \end{bmatrix}$$

$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$ are jointly Gaussian with $m_{\underline{X}} = \underline{0}$ and

$$K_X = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & \frac{1}{2} \end{bmatrix}.$$

X_n, X_{n+m}, X_{n+2m} have the same joint pdf as in part a.

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6.89

(c)

$$X_n = \tfrac{1}{2}U_{n-1} + \tfrac{1}{2}U_n$$

$$Y_n = -\tfrac{1}{2}U_{n-1} + \tfrac{1}{2}U_n$$

Assume $m > n+1$

$$X_m = \tfrac{1}{2}U_{m-1} + \tfrac{1}{2}U_m$$

$$Y_m = -\tfrac{1}{2}U_{m-1} + \tfrac{1}{2}U_m$$

$$\begin{bmatrix} X_n \\ Y_n \\ X_m \\ Y_m \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} U_{n-1} \\ U_n \\ U_{m-1} \\ U_m \end{bmatrix}$$

(X_n, Y_n, X_m, Y_m) are jointly Gaussian with zero mean and

$$K_{\underline{X}} = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

that is, they are iid.

(d)

$$\Phi_{\underline{X}}(\underline{\omega}) = e^{j\underline{\omega}^T\underline{m} - \frac{1}{2}\underline{\omega}^T K \underline{\omega}}$$

a) $$\underline{\omega}^T K \underline{\omega} = (\omega_1, \omega_2, \omega_3) \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{4} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

$$= \tfrac{1}{2}\omega_1^2 + \tfrac{1}{2}\omega_1\omega_2 + \tfrac{1}{2}\omega_2^2 + \tfrac{1}{2}\omega_2\omega_3 + \tfrac{1}{2}\omega_3^2$$

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6.89

(b) $$\omega^T K \omega = (\omega_1, \omega_2, \omega_3)\begin{bmatrix} \frac{1}{2} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

$$= \frac{1}{2}\omega_1^2 - \frac{1}{2}\omega_1\omega_2 + \frac{1}{2}\omega_2^2 - \frac{1}{2}\omega_2\omega_3 + \frac{1}{2}\omega_3^2$$

(c) $$\omega^T K \omega = \frac{1}{2}\omega_1^2 + \frac{1}{2}\omega_2^2 + \frac{1}{2}\omega_3^2 + \frac{1}{2}\omega_4^2$$

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6.90

a) $\hat{X}_2 = aX_1 = bX_3$.

We use the orthogonality principle

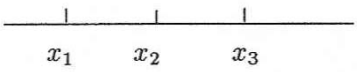
$$\begin{aligned} E[(X_2 - aX_2 - bX_3)X_1] &= 0 \\ E[(X_2 - aX_2 - bX_3)X_3] &= 0 \end{aligned}$$

i.e.

$$\begin{bmatrix} E[X_1^2] & E[X_1X_3] \\ E[X_1X_3] & E[X_3^2] \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} E[X_1X_2] \\ E[X_2X_3] \end{bmatrix}$$

$$\begin{aligned} a &= \frac{VAR[X_3]COV(X_1,X_2) - COV(X_1,X_3)COV(X_2,X_3)}{VAR[X_1]VAR[X_3] - COV(X_1,X_3)^2} \\ b &= \frac{VAR[X_1]COV(X_2,X_3) - COV(X_1,X_2)COV(X_1,X_3)}{VAR[X_1]VAR[X_3] - COV(X_1,X_3)^2} \end{aligned}$$

b)



$$\begin{aligned} a &= \frac{\sigma^2 \cdot \rho_1\sigma^2 - \rho^2\sigma^2 \cdot \rho_1\sigma^2}{\sigma^2 \cdot \sigma^2 - (\rho_2\sigma^2)^2} = \frac{\rho_1 - \rho_1\rho_2}{1 - \rho_2^2} \\ b &= \frac{\sigma^2 \cdot \rho_1\sigma^2 - \rho^1\sigma^2 \cdot \rho_1\sigma^2}{\sigma^2 \cdot \sigma^2 - (\rho_2\sigma^2)^2} = \frac{\rho_1 - \rho_1\rho_2}{1 - \rho_2^2} \end{aligned}$$

$$\begin{aligned} MSE &= E[(X_2 - aX_1 - bX_3)^2] \\ &= E[(X_2 - a(X_1 + X_3))^2] \\ &= E[X_2^2] - 2aE[X_2(X_1 + X_3)] + a^2E[(X_1 + X_3)^2] \\ &= \sigma^2 - 2a(\rho_1\sigma^2 + \rho_1\sigma^2) + a^2(\sigma^2 + 2\rho_2\sigma^2 + \sigma^2) \\ &= \sigma^2[1 - 4a\rho_1 + 2a^2 + 2a^2\rho_2] \\ &= \sigma^2\frac{1 - 2\rho_1^2 + \rho_2}{1 + \rho_2} \\ &= \sigma^2\left(1 - \frac{2\rho_1^2}{1 + \rho_2}\right) . \end{aligned}$$

The interpolation results in smaller MSE.

c) $e = X_2 - aX_1 - bX_3$, e is Gaussian

$$\begin{aligned} E[e] &= 0, \qquad VAR[e] = MSE = \sigma^2(1 - 2\rho_1^2/(1 + \rho_2)) \\ f(e) &= \frac{1}{\sqrt{2\pi VAR[e]}} \exp\left[1\frac{e^2}{2\ VAR[e]}\right] \end{aligned}$$

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6.91 $K = \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix}$

a) If we assume the signals are zero-mean, then the components of $\underline{X}$ correspond to the jointly Gaussian random variables in Ex. 4.13 which are transferred into an independent pair $\underline{Y}$ by the inner transformation given in Ex. 4.36:

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

b) Consider how two consecutive blocks $\underline{X}_1$ are $\underline{X}_2$ are transformed into $\underline{Y}_1$ and $\underline{Y}_2$:

$$\begin{bmatrix} \underline{Y}_1 \\ \underline{Y}_2 \end{bmatrix} = \begin{bmatrix} A\underline{X}_1 \\ A\underline{X}_2 \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

which expanded gives:

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{bmatrix} = \frac{1}{\sqrt{2}} \underbrace{\begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}}_{A'} \begin{bmatrix} X_1 \\ X_2 \\ X-3 \\ X_4 \end{bmatrix}$$

The covariance matrix for $\underline{Y}$ is:

$$AKA^T = \begin{bmatrix} \sigma^2 + \rho\sigma^2 & 0 & \frac{\rho\sigma^2}{2} & -\frac{\rho\sigma^2}{2} \\ 0 & \sigma^2 - \rho\sigma^2 & \frac{\rho\sigma^2}{2} & -\frac{\rho\sigma^2}{2} \\ \frac{\rho\sigma^2}{2} & \frac{\rho\sigma^2}{2} & \sigma^2 + \rho\sigma^2 & 0 \\ -\frac{\rho\sigma^2}{2} & -\frac{\rho\sigma^2}{2} & 0 & \sigma^2 - \rho\sigma^2 \end{bmatrix}$$

It can be seen that the components of $\underline{Y}$ are not independent.

6.92 $X = X_1 + X_2 + ... + X_N$

$$\begin{aligned} m_a &= E[X] = Nm \\ \sigma_N^2 &= VAR[X] = N\ VAR[X_i] = N\sigma^2 \\ f_X(x) &= \frac{1}{\sqrt{2\pi}\sigma_N} \exp\left[-\frac{(x-m_a)^2}{2\sigma_N^2}\right] \end{aligned}$$

a)

$$\begin{aligned} p_{Loss} &= \int_T^\infty f_X(x)dx \\ &= \int_{m_a+t\sigma_N}^\infty \frac{1}{\sqrt{2\pi}\sigma_N} \exp\left(-\frac{(x-m_a)^2}{2\sigma_N^2}\right) dx \\ &= \int_{t\sigma_N}^\infty \frac{1}{\sqrt{2\pi}\sigma_N} \exp\left(-\frac{x^2}{2\sigma_N^2}\right) dx \\ &= \int_t^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx \\ &= Q(t) \end{aligned}$$

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b)
$$\begin{aligned}
E[X_{Loss}] &= \int_T^\infty (x-T)f(x)dx \\
&= \int_T^\infty x\cdot f(x)dx - TQ(t) \\
&= \int_{m_a+t\sigma_N}^\infty x\frac{1}{\sqrt{2\pi}\sigma_N}\exp\left(-\frac{(x-m_a)^2}{2\sigma_N^2}\right)dx - TQ(t) \\
&= \int_{t\sigma_N}^\infty (y+m_a)\frac{1}{\sqrt{2\pi}\sigma_N}\exp\left(-\frac{y^2}{2\sigma_N^2}\right)dy - (m_a+t\sigma_N)Q(t) \\
&= \int_{t\sigma_N}^\infty y\frac{1}{\sqrt{2\pi}\sigma_N}\exp\left(-\frac{y^2}{2\sigma_N^2}\right)dy - t\sigma_N Q(t) \\
&= \int_{\sigma_N}^\infty \sigma_N u\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right)du - t\sigma_N Q(t) \\
&= \sigma_N\int_{\sigma_N}^\infty \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right)d\left(\frac{u^2}{2}\right) - t\sigma_N Q(t) \\
&= \frac{\sigma_N}{\sqrt{2\pi}}e^{-\frac{\sigma_N^2}{2}} - t\sigma_N Q(t)
\end{aligned}$$

c) $\text{fraction of bits lost} = \dfrac{\text{bits lost/33 ms}}{\text{bits produced/33 ms}} = \dfrac{E[X_{Loss}]}{m_a}.$

d)
$$\begin{aligned}
\text{Avg. \# bits allocated per source} &= \frac{m_a+t\sigma_N}{N} \\
&= \frac{Nm+t\sqrt{N}\sigma}{N} = m + t\frac{\sigma}{\sqrt{N}}
\end{aligned}$$

$$\begin{aligned}
\text{Avg. \# bits lost per source} &= \frac{\frac{\sigma_N}{\sqrt{2\pi}}e^{-\frac{\sigma_N^2}{2}} + t\sigma_N Q(t)}{N} \\
&= \frac{\sigma e^{-\sqrt{N}\sigma}}{\sqrt{2\pi N}} + \frac{tQ(t)\sigma}{\sqrt{N}}
\end{aligned}$$

Both quantities decrease with N.

e) We need to keep

$$c = \text{constant} = \frac{E[X_{Loss}]}{m_a} = \frac{\frac{\sqrt{N}\sigma e^{-N\sigma^2/2}}{\sqrt{2\pi}} - t\sqrt{N}\sigma Q(t)}{Nm}$$

$$\begin{aligned}
tQ(t) &= \frac{1}{\sqrt{N}\sigma}\left[\frac{\sqrt{N}\sigma e^{-N\sigma^2/2}}{\sqrt{2\pi}} - Nmc\right] \\
&= \frac{e^{-N\sigma^2/2}}{\sqrt{2\pi}} - \sqrt{N}e\left(\frac{m}{\sigma}\right)
\end{aligned}$$

Solve this equation for t.

f) If $COV(X_i, X_j) = \rho$ then

$$\begin{aligned}
m_a &= Nm \\
\sigma_N^2 &= N\sigma^2 + N(N-1)\rho\sigma^2
\end{aligned}$$

The expressions in terms of m_a and σ_N^2 still hold. However, σ_N^2 now has a stronger dependence on N.

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6.93

(a) $$P[N_1 = j \mid T = t] = \frac{(\lambda_1 t)^j}{j!} e^{-\lambda_1 t}$$

$$P[N_1 = j] = \int_0^\infty \frac{(\lambda_1 t)^j}{j!} e^{-\lambda_1 t} \alpha e^{-\alpha t} \, dt$$

$$= \frac{\alpha \lambda_1^j}{j!} \underbrace{\int_0^\infty t^j e^{-(\alpha+\lambda_1)t} \, dt}_{\frac{j!}{(\alpha+\lambda_1)^{j+1}}}$$

$$= \left(\frac{\alpha}{\alpha+\lambda_1}\right)\left(\frac{\lambda_1}{\alpha+\lambda_1}\right)^j \quad \text{geometric rv } j = 0, 1, \ldots$$

$$f_T[t \mid N = j] = \frac{\frac{(\lambda_1 t)^j}{j!} e^{-\lambda_1 t} \alpha e^{-\alpha t}}{\left(\frac{\alpha}{\alpha+\lambda_1}\right)\left(\frac{\lambda_1}{\alpha+\lambda_1}\right)^j}$$

$$= \frac{(\alpha+\lambda_1)^{j+1}}{j!} t^j e^{-(\alpha+\lambda_1)t}$$

ML $$0 = \frac{d}{dt} P[N_1 = j \mid T = t] = \frac{j(\lambda_1 t)^{j-1}\lambda_1}{j!} e^{-\lambda t} + \frac{(\lambda_1 t)^j}{j!}(-\lambda e^{-\lambda t})$$

$$1 = \frac{\lambda t}{j} \qquad \hat{t} = \frac{j}{\lambda}$$

MAP $$0 = \frac{d}{dt} f[T = t \mid N_1 = j] = c\,\frac{j t^{j-1}}{j!} e^{-(\alpha+\lambda_1)t} + \frac{c\, t^j}{j!}(-\lambda_1 - \alpha) e^{-}$$

$$\Rightarrow \hat{t} = \frac{j}{\alpha+\lambda_1}$$

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(6.93) (b) For linear estimator need covariance

$$E[NT] = E[T\,E[N|T]] = E[\lambda_1 T^2]$$

$$= \lambda_1 E[T^2] = \lambda_1\left[\frac{1}{\alpha^2} + \left(\frac{1}{\alpha}\right)^2\right] = \lambda_1 \frac{2}{\alpha^2}$$

$$E[T] = \frac{1}{\alpha} \qquad E[N] = \frac{\frac{\lambda_1}{\alpha+\lambda}}{\frac{\alpha}{\alpha+\lambda}} = \frac{\lambda_1}{\alpha}$$

$$VAR[T] = \frac{1}{\alpha^2} \qquad VAR[N] = \frac{\frac{\lambda_1}{\alpha+\lambda_1}}{\left(\frac{\alpha}{\alpha+\lambda_1}\right)^2} = \frac{\lambda_1(\alpha+\lambda_1)}{\alpha^2}$$

$$COV(N,T) = \frac{2\lambda_1}{\alpha^2} - \frac{1}{\alpha}\frac{\lambda_1}{\alpha} = \frac{\lambda_1}{\alpha^2}$$

Min MSE Linear Estimator

$$\hat{T} = \frac{COV(N,T)}{VAR[N]}(N - E[N]) + E[T]$$

$$= \frac{\lambda/\alpha^2}{\lambda_1(\alpha+\lambda_1)/\alpha^2}\left(N - \frac{\lambda_1}{\alpha}\right) + \frac{1}{\alpha}$$

$$= \frac{1}{\alpha+\lambda_1} N - \frac{\lambda_1/\alpha}{\alpha+\lambda_1} + \frac{1}{\alpha}$$

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$$P[N_1=j, N_2=k] = \int_0^t \frac{(\lambda_1 t)^j}{j!}\frac{(\lambda_2 t)^k}{k!} e^{-(\lambda_1+\lambda_2)t} \alpha e^{-\alpha t}\, dt$$

$$= \alpha \frac{\lambda_1^j \lambda_2^k}{j!k!} \int_0^\infty t^{j+k} e^{-(\alpha+\lambda_1+\lambda_2)t}\, dt$$

$$\frac{(j+k)!}{(\alpha+\lambda_1+\lambda_2)^{j+k+1}}$$

$$= \binom{j+k}{j}\left(\frac{\alpha}{\alpha+\lambda_1+\lambda_2}\right)\left(\frac{\lambda_1}{\alpha+\lambda_1+\lambda_2}\right)^j\left(\frac{\lambda_2}{\alpha+\lambda_1+\lambda_2}\right)^k$$

$j=0,1,\ldots$

$k=0,1,\ldots$

$$f_T(t|N_1=j,N_2=k) = \frac{\frac{(\lambda_1 t)^j}{j!}\frac{(\lambda_2 t)^k}{k!} e^{-(\lambda_1+\lambda_2)t} \alpha e^{-\alpha t}}{\binom{j+k}{j}\frac{\alpha \lambda_1^j \lambda_2^k}{(\alpha+\lambda_1+\lambda_2)^{j+k+1}}}$$

$$= \frac{(\alpha+\lambda_1+\lambda_2)^{j+k+1} t^{j+k} e^{-(\alpha+\lambda_1+\lambda_2)t}}{(j+k)!}$$

Same basic form as part (a).

Following same procedure we find

$$\hat{T}_{ML} = \frac{j+k}{(\lambda_1+\lambda_2)} \qquad \hat{T}_{MAP} = \frac{j+k}{(\alpha+\lambda_1+\lambda_2)}$$

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